

TruthLens: A Verification-Gated Pipeline for Evidence-Grounded Fact-Checking of Short-Form Political Video

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Abstract—Short-form political video and image posts are now a primary vector for misinformation, yet most automated fact-checking systems are evaluated on whether they reach a correct final label, not on whether the evidence chain that produced it is real, and few are evaluated against independent, held-out ground truth at all. We present TruthLens, a verification-gated pipeline that decomposes a post into atomic claims, researches each against the open web, and subjects every LLM-generated verdict to a deterministic, non-LLM validator before publication – checking citation existence, source-fetch existence, numeric grounding, and label/reasoning consistency. This paper makes three contributions. (1) A methodological correction, reported with the same prominence as any positive result. A skeptical post-hoc audit of our own experimental design found that, in every run across this project, the baselines had been given a clean, human-written claim summary rather than the claim TruthLens’s own extraction stage produced – an undisclosed confound that inflated baseline performance relative to our own system. Corrected, with claim extraction held constant across all configurations, full TruthLens (33.3%, 2/6) beats two of three baselines outright (0.0% each) and trails the third by 16.7 points, down from a 33.4-point gap, on the paired six-item subset where every configuration produced a verdict. (2) A construct and an evaluation of it. We formally define support validity – whether a verdict’s citations, sources, numbers, and label/reasoning consistency hold up against its own evidence matrix – and show that two ground-truth-independent validator fixes raise validator recall (16.7% $\rightarrow$ 40%, 100% precision throughout) without moving reel-level accuracy at all: evidence that label accuracy and support validity are separable, independently-moving properties, and the more informative axis for an architecture whose premise is evidence-before-verdict. A claim-decomposition counterfactual ($n = 4$) separately isolates deterministic aggregation as a real contributor (50.0% vs. 25.0% accuracy) once claim extraction has more than one claim to work with. (3) A held-out evaluation benchmark. A measured structural finding – that sourcing this misinformation pattern from Instagram alone yields roughly one usable item per 900 candidates screened, because the pattern predominantly recirculates on X/Twitter – led us to relax the platform constraint and build TruthLens-202, a 202-item held-out benchmark of real short-form political posts (X, Instagram, Facebook), each anchored to an independent professional fact-check via a verbatim evidence quote under a review-gated annotation protocol, with 206 documented same-media cross-post relationships. We report its full composition, including three domain-driven skews (83% FALSE, 80% X-sourced, 80% provenance-type claims) and a source concentration ($\sim$ 88% Alt News). We then run the complete pipeline over all 193 validation items in a \$0, no-API-key, fully local configuration (every stage on a local 3B model,

keyless search, no cloud escalation). The result is deliberately unflattering and is reported as a floor, not a headline: only 34.7% of items produce a verdict at all, and on those the bucketed accuracy is 29.9% (Wilson 95% CI [20.2, 41.7]), below an always-FALSE baseline’s balanced accuracy. The binding constraint is localised precisely – claim extraction, not evidence retrieval (0% research-failure) or verdict reasoning: 65.3% of real posts yield no checkable claim to the local model, and the MISLEADING class is never once recovered (0/7). Strengthening that configuration, and a second independent annotator pass, are the stated next steps. Alongside these we report the evidence-quality picture behind the frozen results (a primary-source domain-restriction fix moving source-tier classification from $\sim$ 11% to 95% of retrieved sources; a still-open gap between finding topically-relevant primary sources (68.75%) and extracting usable evidence from them (18.75%); a bug found mid-evaluation in which 47.1% of evidence rows carried an empty explanation field), a multimodal claim-coverage experiment in which OCR-derived evidence recovered a claim audio transcription missed entirely, a live-tested prompt-injection defense (1/5 pre-fix, 0/5 post-fix, with the downstream validator independently neutralizing the pre-fix success), a newly-found and honestly-unresolved verdict-stage failure (reliability weighting under direct source contradiction; 0/14 trials before and after a targeted fix), and honestly-scoped non-results for political-bias and calibration questions this data’s size cannot yet support. A 24-entry taxonomy of failure modes surfaced by running the complete system against real content – over a third found during the evaluation program itself, all but three with a deterministic check and a regression test built from the case that surfaced it – is as practically useful a contribution as any single number above. Every number is reported with its exact sample size and, where a proportion, a Wilson confidence interval; where an experiment could not be run at a scientifically meaningful scale, or where our own methodology was found to be flawed, we say so rather than force or hide it.

Index Terms—automated fact-checking, large language models, evidence grounding, retrieval-augmented generation, misinformation detection, short-form video, claim verification, source credibility, experimental methodology

I. Introduction

Political misinformation increasingly spreads through short-form video – Instagram Reels, YouTube Shorts, TikTok – rather than text [1]. This shift matters for automated fact-checking research in two ways. First, the input is multi-modal and lossy: a claim may be spoken (transcript), displayed as on-screen text (OCR), or only

implied by a caption, and a system that treats a reel as a single block of text discards exactly the structure a fact-checker needs. Second, and more consequentially, the dominant evaluation paradigm – did the system output the correct veracity label, measured against a benchmark such as FEVER [2] or LIAR [3] – says nothing about whether the evidence a system cites is real, or whether a correct label was reached by checking the actual claim being made rather than a different, tangential one. An LLM asked to fact-check a claim can produce a fluent, well-cited verdict whose citations point to sources it never retrieved, whose numbers never appeared in any source it read, or whose confident label contradicts its own stated reasoning.

We built TruthLens to make that failure mode structurally difficult rather than merely discouraged by prompting. TruthLens ingests a political short-form video (or photo post), decomposes it into atomic factual claims, researches each against the open web through a tiered-source retrieval process, and only then proposes a claim-level verdict – a hard evidence-before-verdict ordering enforced architecturally, not just by instruction (Section IV). Every LLM-generated verdict is additionally subject to a deterministic, non-LLM validation layer before it can be published, and every pipeline stage escalates from a free, locally hosted model to a paid one only when a deterministic quality gate specific to that stage’s own failure modes fails.

This paper reports the first evaluation of TruthLens against independent, held-out ground truth, rather than development-time telemetry alone. We curated a frozen 9-item comparison set of real Instagram posts, each independently fact-checked by a professional Indian fact-checking organization (BOOM Live, Alt News, Factly), disjoint from every post used during development (Section VI), and compared TruthLens against three baselines that hold the underlying model and search infrastructure constant, isolating what TruthLens’s multi-stage architecture adds beyond raw search access (Section V). Separately, and after that frozen comparison, we release TruthLens-202: a 202-item held-out corpus of real short-form political posts across X, Instagram and Facebook, built once a measured structural finding showed Instagram-only sourcing to be capped at roughly one usable item per 900 candidates screened. TruthLens-202’s construction and limitations are documented in full (Section VI-F), and we report the first end-to-end run over its 193 validation items in a \$0 local-only configuration (Section VII-D): a deliberate floor, on which 34.7% of posts reach a verdict and bucketed accuracy is 29.9%, with claim extraction – not retrieval or verdict reasoning – identified as the component that decides whether a post gets checked at all. We report every result this comparison produced, including one that is not flattering but not about TruthLens: a subsequent, skeptical audit of our own experimental design found that the baselines had been given a cleaner claim input

than TruthLens itself ever worked with, contradicting our own stated methodology, undetected until this audit (Section VII-A). We report the bug and its fix with the same prominence as any other finding, because a flaw that inflates a comparison against our own system is exactly what our own protocol requires us to surface rather than quietly correct. Once corrected, full TruthLens beats two of the three baselines outright and trails the third by a substantially smaller margin than the uncorrected comparison showed (Section VII) – and a companion counterfactual analysis of claim decomposition shows the same architecture’s aggregation mechanism demonstrably helps once claim extraction has more than one claim to work with (Section VIII). We treat the correction, the corrected result, and the ablation as more scientifically useful, together, than any single headline number on its own.

This work is organized around six research questions, matching TruthLens’s own architectural claims one-to-one so each is either supported by a specific experiment or explicitly withdrawn: whether deterministic verification reduces unsupported output (RQ1); whether multi-stage decomposition and retrieval beat single-shot search (RQ2); whether multimodal extraction improves coverage of non-textual misinformation (RQ3); whether source-tiering improves evidence quality (RQ4); whether the system maintains comparable standards across political actors (RQ5); and whether its stated confidence correlates with correctness (RQ6). Section III states these formally alongside this paper’s specific, experimentally-supported contributions.

We are explicit throughout about the evidentiary status of every claim: 9 held-out items (6 with a resolved verdict from every configuration), single-annotator Tier-2 judgment where used, and RQ5/RQ6 explicitly deferred rather than forced past what this sample size can support (Section XI). Sections XIV and XVI state what would need to change before any number here should be read as a stable, generalizable estimate.

II. Related Work

Automated fact-checking and LLM factuality evaluation have both grown into large enough literatures to warrant their own surveys [4], [5]. None of the ideas below – retrieval-augmented LLM fact-checking, claim decomposition, source-tiering, or multimodal misinformation detection – originate with this work; our contribution (Section III) is a specific combination of them gated by deterministic verification, evaluated honestly including where it does not yet outperform simpler alternatives. Extended per-subsection discussion, including how our headline result reads against DeVerna et al.’s finding below, is in the Discussion (Section XV).

Claim detection and decomposition. ClaimBuster [6] ranks existing sentences by check-worthiness via a TF-IDF/POS/NER SVM; TruthLens’s claim-extraction stage

instead decomposes a transcript/OCR/caption into new atomic claims, with a grounding requirement specific to video: each claim’s `source_quote`, when present, must be a verbatim substring of the actual transcript/OCR text, checked deterministically. Section VIII gives the first controlled measurement of whether this decomposition improves downstream verdict correctness.

Fact-checking benchmarks. FEVER [2] and LIAR [3] are the dominant claim-verification benchmarks; both evaluate a final label against ground truth without examining whether the evidence chain that produced it is real. TruthLens’s own 9-item benchmark (Section VI) is a modest step toward the same kind of independent evaluation for our domain – three orders of magnitude smaller, sourced from real Indian professional fact-checks of Instagram posts rather than curated text.

LLM hallucination and grounding verification. Commercial legal AI tools hallucinate on 17–33% of queries by hand-scored evaluation [7]; CiteCheck [8] detects hallucinated references via a structured LLM verifier. TruthLens’s validation layer shares the goal but uses deterministic set-membership/substring tests instead, trading generality for auditability and zero marginal LLM cost – a tradeoff whose limits (cannot catch reasoning phrased outside observed patterns) are measured, not assumed, in Section IX. A legal-domain citation-grounding-oracle analysis is a relevant caution: apparent hallucination rates are highly sensitive to reference-graph coverage, not just the model measured [9] – TruthLens’s own number-grounding check has an analogous blind spot. Per Rule 8 of this program’s own protocol, we never call a rate a “hallucination rate” without human annotation behind it, using grounding-constraint violation rate and unsupported generation rate instead.

Political fact-checking with LLMs. DeVerna et al. [10] found reasoning ability provides “minimal benefit” and raw search only “moderate gains” for political fact-checking, while curated context improved macro-F1 by 233% across 15 LLMs – strong independent evidence for TruthLens’s core design bet that evidence curation matters more than final-step reasoning. Our own corrected result (Section VII) is broadly consistent with this once a real baseline-design confound is removed (Discussion, Section XV).

Short-form video misinformation. TikTec [1] classifies misleading COVID-19 TikTok videos via joint visual/audio/text modeling; recent work targets multilingual short-form-video checkworthiness specifically [11]. TruthLens differs in task framing (open-domain, claim-level, cited verification, closer to FEVER-style verification than TikTec-style classification) and, per Section X-E, shares this literature’s blind spot to genuine footage recirculated under a new caption.

Political neutrality of automated fact-checkers. Frontier LLMs show measurable partisan asymmetries under politically charged prompts [12]; every TruthLens system

prompt includes an explicit neutrality clause, but RQ5’s quantitative bias evaluation could not be run at a meaningful sample size (Section XI), reported as such rather than skipped or forced.

Cost-aware model cascading. FrugalGPT [13] routes most queries to a cheap model, escalating hard cases to an expensive one. TruthLens applies the same principle in reverse-priority form: every stage defaults to a free, local model, escalating only when a deterministic quality gate – not a learned predictor – detects substantively deficient output (Section IV-D2).

A. Positioning against related systems

Table I summarizes where TruthLens sits relative to the systems discussed above, on dimensions each subsection has already individually justified. We include only systems whose relevant capabilities are established by the citations already discussed in this section, to avoid characterizing a system’s design beyond what we have verified. TruthLens is the only row combining multimodal input, claim decomposition, source tiering, deterministic (not LLM-judged) output verification, and held-out professional fact-check ground truth – combining these is TruthLens’s contribution, as Section III states explicitly; no individual capability in this table originates with TruthLens.

Stated plainly, per this paper’s own governing discipline against overclaiming: TruthLens claims novelty for no single column in Table I – multimodal short-form-video analysis predates it (TikTec), political-domain fact-checking with LLM-mediated retrieval predates it (DeVerna et al.), and citation grounding against an external source predates it, including via a non-deterministic LLM-judged mechanism this paper argues against on cost and auditability grounds, not the underlying goal (CiteCheck). What we have not found is a system combining held-out professional-fact-check evaluation, multimodal claim extraction, source-tiered retrieval, and deterministic output verification together, evaluated on real, recirculated short-form political video – the specific gap Sections IV–XI target and evaluate honestly, including where the combination does not yet outperform a simpler alternative (Section VII).

III. Research Questions and Contributions

Table II states the six research questions this program was designed against, where each is answered, and its evidentiary status, following Rule 12 of our own governing protocol: an RQ that cannot be evaluated at a meaningful sample size is reported as deferred, not silently dropped or forced.

This paper makes the following contributions, limited to what the experiments in Sections VII–XI actually support:

- 1) A held-out, Tier-1 evaluation benchmark and controlled baseline/counterfactual comparison (Sections V–VI), the first evaluation against independent professional ground truth rather than

TABLE I
Positioning against related systems (✓ = present, ✗ = absent, ? = not established by the citation discussed)

System	Multimodal	Short-video	Political	Claim decomp.	Retrieval/RAG	Source tiering	Verification
FEVER [2]	✗	✗	✗	✗	✓	✗	none (label-only benchmark)
LIAR [3]	✗	✗	✓	✗	✗	✗	none (label-only benchmark)
ClaimBuster [6]	✗	✗	✓	✗	✗	✗	n/a (checkworthiness only)
TikTec [1]	✓	✓	✗	✗	✗	✗	none (classifier output)
DeVerna et al. [10]	✗	✗	✓	✗	✓	✗	none (studies LLM judges)
CiteCheck [8]	✗	✗	✗	✗	?	✗	LLM-judged citation check
TruthLens (this paper)	✓	✓	✓	✓	✓	✓	deterministic, 4-check

TABLE II
Research questions, where answered, and status

RQ	Question	Status
1	Does deterministic post-hoc validation reduce unsupported verdict output?	Yes, directional (§IX, $n = 9$)
2	Does multi-stage decomposition + retrieval beat single-shot search+LLM?	Partially, once claim input is corrected: beats 2 of 3 baselines, trails the 3rd (§VII); yes for aggregation specifically (§VIII, $n = 4$)
3	Does multimodal extraction improve coverage of non-textual misinformation?	Directional yes (§X, $n = 6$)
4	Does source-tiering improve evidence quality?	Yes for tier classification; gap remains to usable evidence (§IX)
5	Comparable standards across political actors?	Deferred – no matched pairs exist (§XI)
6	Does confidence correlate with correctness?	Deferred – sample too small to calibrate (§XI)

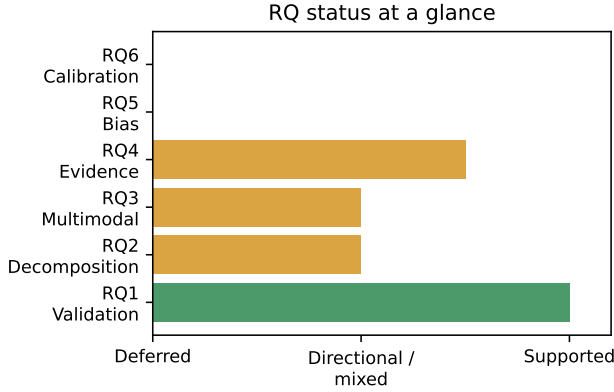


Fig. 1. Status of each research question at a glance (green = supported, amber = directional/mixed, red = deferred). Derived from Table II; not new data.

development-time telemetry, holding model, search infrastructure, and (Section VII-A) claim-extraction input constant so measured differences are attributable to downstream architecture, not to which configuration had an easier claim to reason about.

- 2) A documented, fixed methodological confound in our own baseline design, reported with the same prominence as any other finding. Baselines 1–3 had, in every run across this project, been given a clean, human-written claim summary rather than TruthLens’s own extracted claim (Section VII-A), contradicting our stated methodology and inflating baseline performance. We fixed it and re-ran the comparison rather than silently correcting the number, reversing this program’s headline result in the

process.

- 3) An empirical demonstration that verdict-label accuracy and support validity are separable, independently-moving properties (Section IX). Two general, ground-truth-independent validator fixes improved recall (16.7%→40%, $n = 9$, 100% precision throughout) while leaving reel-level accuracy unchanged (2/6 before and after) – outputs became measurably more supported by their own cited evidence without becoming more “accurate” by the bucket-matching standard.
- 4) A claim-decomposition counterfactual analysis isolating the clearest component-level result in this program, where aggregation demonstrably outperformed a simpler alternative (50.0% vs. 25.0%, $n = 4$, Section VIII).
- 5) A four-way evidence-quality analysis separating source -tier classification, topical relevance, fetch success, and usable-evidence extraction (Section IX): retrieval finds real, on-topic primary sources at a meaningfully higher rate (68.75%) than usable-evidence rate (18.75%) would suggest – the bottleneck is source specificity, not retrieval relevance.
- 6) A cross-referenced taxonomy of real failure modes (Section XIII), each found by running the complete system against real content end-to-end and fixed with a deterministic check and a regression test built from the case that surfaced it.
- 7) TruthLens-202: a 202-item held-out benchmark of real short-form political posts (X, Instagram, Facebook), each anchored to an independent professional fact-check via a verbatim evidence quote under a review-gated protocol, with 206 documented same-

media cross-post relationships (Section VI-F). Its full composition and limitations – a single LLM-assisted reviewer, $\sim 88\%$ single-source ground truth, and three domain-driven skews – are reported in full. We also contribute the first end-to-end result on it (Section VII-D): a local-only, \$0 configuration run over all 193 validation items, reported as a floor – 34.7% of posts reach a verdict, 29.9% bucketed accuracy on those, and claim extraction identified as the binding constraint (65.3% no-checkable-claim rate at 0% infrastructure failure).

TruthLens does not claim to have invented retrieval-augmented LLM fact-checking, claim decomposition, source-credibility tiering, or multimodal misinformation detection – each independently pre-dates this work, as Section II discusses. Its contribution is the specific combination of these mechanisms gated by deterministic, non-LLM verification, evaluated against independent ground truth including where that combination does not yet outperform a simpler alternative.

IV. TruthLens Architecture

TruthLens is a FastAPI backend with a PostgreSQL/pgvector datastore, a Next.js review dashboard, and an object store (S3-compatible) for media and rendered output. Figure 2 shows the 9-stage pipeline; no stage begins before the previous one has persisted its output. Stage 1 (ingestion) downloads or accepts a video or photo, transcribes audio, OCRs on-screen text at a fixed interval, and selects a thumbnail (Section IV-C covers the photo-post path). Stage 2 (claim extraction) has an LLM decompose the transcript/OCR/ caption into atomic claims classified factual/opinion/ prediction/satire/rhetorical, with only specific factual claims marked verifiable. Stage 3 (research planning) proposes 2–5 targeted search queries per verifiable claim, explicitly requesting at least one primary-source-targeted query. Stage 4 (search and fetch) runs each query and extracts full page text, not just the snippet. Stage 5 (evidence analysis) has an LLM judge stance (supports/contradicts/provides context/ irrelevant) per fetched source, grounded only in that source’s own extracted text. Stage 6 (verdict proposal) gives an LLM the claim and its full evidence matrix and asks for a label, confidence, and cited evidence IDs. Stage 7 is deterministic validation (Section IV-D1). Stage 8 derives one reel-level verdict from a fixed rule table over already-validated per-claim verdicts, not a further LLM call. Stage 9 assembles a carousel from already-validated database rows, with only headline phrasing and a short “why” paragraph passing through one additional narrow, grounded LLM call each (Section IV-D2).

A. Source tiering

Every retrieved source is scored along an 8-dimension weighted reliability rubric and assigned to one of

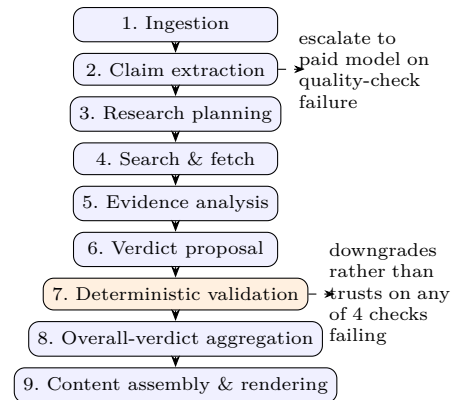


Fig. 2. The 9-stage TruthLens pipeline (Section IV). No stage begins before the one before it has persisted its output. Stage 7 (orange) is the deterministic, non-LLM validation gate (Section IV-D1); five stages (2, 5, 6, and the two content -generation calls inside stage 9, plus stage 5’s newer sibling check) run their own substantiveness check before accepting local-model output (Section IV-D2).

eight tiers: `primary_government`, `primary_legal`, `primary_data`, `news_wire`, `established_news`, `academic`, `factcheck_org`, or other. Tier membership determines both which sources are eligible to be surfaced as a claim’s “primary source” citation on the rendered carousel and, in aggregate, the reliability-weighted confidence of the evidence matrix handed to the verdict-proposal stage. RQ4 (Section IX) evaluates this mechanism directly. The eight dimensions’ weights are hand-set manual priors, not derived from any formal method or learned fit – verified via git history to be unmodified since before any evaluation dataset existed. A sensitivity analysis against an unweighted or alternative rubric is unperformed, named in Section XVI.

B. Multi-claim, reel-level verdicts

TruthLens builds one fact-check per reel, covering every verifiable claim extracted from it – collapsing several claims of differing veracity into one verdict discards exactly the “partially true, partially misleading” structure a reader needs. A fixed rule table combines per-claim verdicts into the published carousel’s one verdict (Section IV-D3); Section VIII tests whether this multi-claim structure improves accuracy over the single highest-importance claim alone.

C. Multi-modal ingestion: video and photo posts

When the video downloader reports no video stream, the pipeline falls back to the post’s Open Graph metadata (image URL, caption, engagement counts) fetched directly from the page. A `media_type` field drives branching at exactly one point: audio/transcription are skipped for a photo, but OCR and vision-context needed no new code, since both already operate on a plain list of image paths – a photo is a one-element case of the same interface video frames already satisfy. Section X evaluates claim coverage across this system’s three real input-modality conditions.

D. Making the model’s output more supportable

We make precise, in place of the vaguer word “trustworthy,” a single construct used throughout this paper: a verdict has support validity to the extent that every specific, checkable assertion in it – cited evidence IDs, the sources they point to, reasoning numbers, and the logical relationship between reasoning and label – can be verified against data already in the pipeline’s evidence matrix, without a second model judgment. Support validity is deliberately narrower than, and does not imply, label accuracy or general absence of hallucination: a verdict can have perfect support validity and still be wrong, if the model correctly cited evidence it misinterpreted, or if retrieval never found the right evidence at all (Section IX’s four-way metric addresses that). Sections VII–XI evaluate whether the mechanisms below actually make this checkable.

1) Deterministic evidence-grounding validation: After the verdict-proposal LLM call returns a structured VerdictProposal (label, confidence, reasoning, cited evidence IDs), TruthLens runs four checks against data already in the database, none of which invoke an LLM:

- Citation existence: every ID in `cited_evidence_ids` must correspond to an evidence row that was actually passed to the model in its evidence matrix.
- Source-fetch existence: every cited evidence row’s source must have a stored, successfully fetched full-text passage – a citation cannot point at a source the pipeline only planned to fetch.
- Number grounding: every numeral appearing in the model’s `reasoning_summary` is extracted and checked for appearance in at least one cited source’s stored passage text. UUID-shaped hex sequences (from internal citation markup) are stripped before number extraction – a real bug, described in Section XIII, in which digits from a cited evidence ID’s own UUID were being extracted as apparent “unsupported statistics.”
- Reasoning/label consistency (new). A verdict whose own `reasoning_summary` states that no reliable evidence was found (matched against a fixed phrase list, e.g. “does not provide any reliable,” “no reliable sources”) but whose label is a confident value other than UNVERIFIED is downgraded to UNVERIFIED. Added in direct response to Section IX’s finding that this specific pattern was the largest single category of validator false negative; its circularity as a threat to validity is disclosed, not hidden, in Section IX.

These four checks are the entire current operationalization of support validity – deliberately not a proxy for evidence-to-claim topical relevance (Section IX’s job), evidence-to-reasoning stance correctness (human-audited only, a named gap), or entity consistency (not yet implemented, Section XVI). A verdict that fails any check is downgraded, not discarded: persisted with a non-passed

`validation_status`, an appended [VALIDATION NOTE: ...], and excluded from citation lists.

UNVERIFIED means the system genuinely searched and found insufficient evidence; a distinct RESEARCH_FAILED status is raised only when all queries fail at the infrastructure level, and a publication gate refuses to build a fact-check from it – distinguishing “I don’t know” from “I could not check.”

2) Verification-gated model cascading: Every stage defaults to a free, local model. Five stages run a deterministic substantiveness check before accepting local-model output – e.g. vision-context requires its `scene_description` to clear a 0.20 word-overlap-with-prompt threshold (calibrated against every real garbled/coherent example collected: garbled 0.26–0.29, coherent 0.00–0.12), and evidence analysis requires a non-empty per-source explanation (added after Section IX found 47.1% empty). On failure, the call retries once against a paid model (Gemini) with an identical prompt – unlike a classic cascade [13], the trigger is a deterministic per-stage predicate, not a learned confidence estimate. Full per-stage thresholds and illustrative firing-rate telemetry: Appendix A.

3) Deterministic aggregation and grounded corrections: Once every claim has a verdict, TruthLens derives one reel-level verdict via a fixed rule table over the set of per-claim labels (weighted by importance ≥ 0.5 as “major”): a plain combinatorial function, not a further LLM call, keeping the published carousel’s headline verdict auditable from already-validated per-claim verdicts (Section VIII evaluates this mechanism directly). Optional corrected `fact/context_note` fields, held to the same evidence-grounding rule, are also produced and invalidated alongside the parent verdict on any validation failure; full rule table and field details: Appendix A.

E. Stage 10: human-approval-gated publishing

Stage 9’s rendered carousel is not posted automatically. A tenth stage, outside the 9-stage accuracy pipeline Figure 2 diagrams and outside every research question this paper evaluates, publishes an approved fact-check to a real Instagram account via the official Meta Graph API’s Content Publishing endpoints – no browser automation or unofficial API of any kind, matching this project’s stated Terms-of-Service-compliance posture (docs/API_REQUIREMENTS.md §7). The state machine (`researching`→`ready_for_review`→`approved`→`published`) enforces a human decision at `ready_for_review`: nothing reaches Instagram without an explicit reviewer approval, and approval and publish are two separate API calls against two separate database rows (a `GeneratedPost` created at approval time, a `PublishingJob` created at publish time) rather than one combined action, specifically so a failed Graph API call cannot silently re-trigger a duplicate approval or a double post. The publish call itself is a fixed six-step sequence per Instagram’s own documented protocol: create one

media container per slide, poll each until Instagram’s servers report FINISHED, create a carousel container referencing all four, poll it, then publish and retrieve the permalink – each step individually retried (three attempts, exponential backoff) but the overall sequence not idempotent, which is what the approval/publish split above is defending against. Cost is \$0 (Meta’s standard platform terms; the observed constraint is a rate limit, not a price – roughly 25 media_publish calls per 24 hours per account, against which this project’s own configured cap of 12 posts/day leaves real but not lavish margin).

This stage has existed since the project’s initial build but, unlike every accuracy-pipeline stage above, had zero test coverage before this pass – a real gap for the one piece of code with an external, real-world, hard-to-fully-undo side effect. 18 tests were added: the full six-step Graph API sequence against a mocked HTTP boundary (container creation, ready-polling including a timeout case, carousel creation, publish, permalink retrieval), both real error shapes the Graph API actually returns (an HTTP error status, and a 200 response carrying an error object), the 2–10 item carousel-size guard, and – against a real row graph, not mocks, for everything except the Graph API call itself – the already-published refusal, the missing-access-token refusal, and that a Graph API failure marks the publishing job failed with a real stored error rather than leaving it stuck or incorrectly marking the fact-check published anyway. No test exercises a live Instagram account: doing so requires the operator to complete a one-time account-linking step (an Instagram Business account linked to a Facebook Page, added as a Tester on a Meta developer app – free, but outside this codebase, documented in docs/API_REQUIREMENTS.md §1) that had not been completed as of this writing (Section XVII).

V. Experimental Methodology

A. Central research question and design

The central question motivating this evaluation: can a verification-gated, evidence-grounded architecture reduce unsupported fact-checking outputs while preserving or improving factual correctness in short-form political media? We answer this via a controlled comparison across five system configurations, all run against the same underlying local model (llama3.2 via Ollama) and, where the configuration has search access at all, the same search provider (DuckDuckGo) TruthLens itself uses in production – holding model and search-infrastructure quality constant so that a measured difference is attributable to architecture, not to a cheap-model-vs.-expensive-model confound.

B. System configurations

Five configurations, all on llama3.2: B1 (LLM-only) – claim → verdict, no search/evidence/validation. B2 (Search+LLM) – claim → one DuckDuckGo search (top-5 snippets, no full-page fetch) → verdict; the claim is passed to search verbatim, no research-planning stage, the single

most important difference from B3 and TruthLens. B3 (Search+RAG+LLM) – adds full-page fetch/ extraction over the same results, isolating retrieval depth from decomposition and per-source analysis. B4 (TruthLens minus validation) – the real pipeline with validate_verdict()’s outcome recorded but never applied (a runtime flag, SKIP_VALIDATION, not a forked codepath), the primary RQ1 ablation. Full TruthLens – unmodified, at the frozen evaluation tag. Decomposition, source tiering, per-source analysis, and validation exist only in the full system; Baselines 1–3 operate on TruthLens’s own already-extracted claim, since claim extraction itself is RQ3’s job, not RQ2’s. A real infrastructure bug was found and fixed while building B2/B3 (the first draft silently inherited TruthLens’s own Gemini-fallback mechanism via a shared provider factory, caught during smoke testing; Section XIII).

C. Ablations, metrics, and held-out discipline

Beyond B4 (RQ1), two further ablations: a source-tier ablation (domain-restricted vs. unrestricted retrieval, Section IX) and a claim-decomposition ablation (Section VIII), computed from already-collected verdict data at zero additional LLM cost. Every proportion is reported with its exact numerator/denominator and a 95% Wilson CI; paired comparisons use McNemar’s exact test with the raw contingency table, never a bare p -value; accuracy uses a fixed VerdictLabel-to-ground -truth bucketing where UNVERIFIED never counts as a match. Full formulas are fixed in advance in METRICS.md and not altered after seeing results. The dataset (Section VI) is checked item-by-item against already-ingested development content and excluded on any match; no prompt, threshold, or model choice was changed based on dataset contents or performance after an item was added – the two general fixes in Section IX are deliberately ground-truth-independent, with that distinction’s limits argued explicitly, not asserted. Two tagged freezes (dataset before Baselines 1–3; system code before the held-out run) mean any later code change is logged, never silently applied.

D. Honest scope reductions

Stated rather than hidden: 9 items, not 50–100 (a pilot found ~8% hit rate for a usable Tier-1 candidate; 20–30 items would need 250–375 more articles checked, Section VI). Single annotator, no IAA statistic – Tier-1 ground truth is a professional organization’s own verdict, authored by no one on this project; any Tier-2 judgment used (draft claim-coverage/evidence -relevance labels only, never a headline verdict) is explicitly marked as such. RQ3, RQ5, RQ6 reported as directional or deferred, not generalizable rates. No p -values from tests assuming a sample size this program does not reach – Wilson intervals and McNemar’s exact test only, reported as uncertainty descriptions, not significance claims.

VI. Dataset and Annotation

This work uses two disjoint held-out sets. The 9-item comparison set (Sections VI.B–F) is the frozen dev split against which every controlled result in Section VII and the foundation-phase experiments (Section XII) are computed; it is small by deliberate, disclosed choice (Section V-D) and is never re-touched. TruthLens-202 (Section VI-F) is a separate, larger validation split, built after the frozen comparison, that the platform-scope and sourcing findings of this program made feasible. It is evaluated end-to-end once here, in a single local-only configuration (Section VII-D); that number is reported as a floor and is kept strictly separate from the frozen dev-split results, which it does not affect.

A. Ground-truth tiers

Tier 1 (used for every item in this evaluation): the claim in the Instagram post has already been verdicted by an independent professional fact-checking organization, and that organization’s article demonstrably references or embeds this specific post, not merely a similar claim elsewhere. Ground truth is their published verdict, with the article URL stored for auditability. Tier 2 (specified, not used for any headline verdict in this evaluation): no professional fact-check exists; a project annotator labels directly from primary sources under a documented protocol. No item is ever labeled by TruthLens itself, or by any LLM, and then used as ground truth.

B. The 9-item benchmark

Table III lists every item. All 9 are Tier 1, sourced from BOOM Live, Alt News, and Factly. item-0003 (a CJP-investigated, AI-generated visual claim) could not be ingested after three independent attempts across this program (a persistent Instagram-side “empty media response”), and is excluded from every table in this paper, not scored as a failure.

C. Sourcing method and hit rate

Items were sourced by searching professional fact-checking outlets for articles that embed a still-live Instagram post, prioritizing coverage gaps in claim type, language, and political actor across three sourcing passes. The method’s real, load-bearing constraint is a $\sim 8\%$ hit rate (2 of the first 26 fact-check articles checked yielded a usable, still-live, directly-embedded post) – most professional fact-checks either do not embed the specific post, or it has since been taken down. Every accepted item’s content hash is checked against already-ingested development data and excluded on any match.

D. Inter-annotator agreement

No IAA statistic is computed, and none is fabricated to fill this gap. Tier-1 ground truth does not have an “annotator” in the IAA sense – the label is a single organization’s single published verdict, and introducing

a second in-project annotator to re-judge it would defeat the purpose of using independent ground truth. No Tier-2 items exist in the frozen dataset, so the condition under which IAA would be meaningful does not currently arise; if Tier-2 items are added with only one available annotator, this will continue to be reported as “single-annotator ground truth, no IAA computed,” not papered over with an unearned kappa statistic.

E. Composition gaps, disclosed

Five of 9 items are provenance-type claims; statistic, law_policy, historical, and true_claim remain unrepresented after three sourcing passes – plausibly, though unprovenly, a property of which claim types concentrate on Instagram versus other platforms in this domain. Seven of 9 items are FALSE, likely reflecting that fact-checkers publish far more debunks than confirmations; only item-0004 is TRUE. Two items (item-0007, item-0008) have confirmed Hindi/mixed spoken content, still a thin representation of the project’s stated multilingual target. BJP has 3 items (the most of any actor, alongside 6 single-item actors), but none form a controlled topic/structure/difficulty match against another actor, which is why RQ5 (Section XI) is deferred rather than attempted at $n = 1$ -per-comparison.

F. TruthLens-202: a larger held-out benchmark

The 9-item comparison set above is frozen and never expanded. Separately, this program built a larger held-out corpus, TruthLens-202, for end-to-end evaluation. It comprises 202 real short-form political posts: the 9 comparison items form its dev split, and 193 newly collected items form a validation split, evaluated once in this paper (local-only configuration, Section VII-D) and kept separate from the frozen dev-split results. We describe how it was built, its composition, and its limitations in full, because the way an evaluation set is constructed is itself a claim that should be checkable.

1) Why the platform scope was relaxed: The five earlier sourcing passes (Sections VI, XII, XII-F) restricted candidates to Instagram. Section XII-F measured the cost of that restriction directly: across roughly 3,600 candidates screened over five Indian fact-checker archives, the blended yield of a usable Instagram-native misinformation-source post was on the order of one per 900, an order of magnitude below the original manual-sourcing estimate. That finding, first observed qualitatively in Section X (most of this pattern’s false claims are attached on X/Twitter, not Instagram) and then confirmed quantitatively at volume, is what motivated relaxing the constraint: not a preference for scale over rigor, but a measured structural property of where this misinformation actually recirculates. TruthLens-202’s validation split therefore admits posts from any platform whose media is retrievable through an official/sanctioned path (X/Twitter, Facebook, Instagram, and in principle YouTube and TikTok), while holding

TABLE III
Held-out benchmark: 9 Tier-1 items (all sourced independently of TruthLens or its developer)

ID	Ground truth	Source	Political actor	Claim type
item-0001	FALSE	BOOM Live	BJP	provenance
item-0002	FALSE	Alt News	Karni Sena	provenance (visual)
item-0003 [†]	FALSE	BOOM Live	CJP	visual (AI-generated)
item-0004	TRUE	BOOM Live	Delhi Police (government)	provenance
item-0005	MISLEADING	Factly	Congress	provenance
item-0006	FALSE	BOOM Live	none (disaster misinfo)	event (visual)
item-0007	FALSE	Alt News	BJP	misleading context
item-0008	FALSE	Alt News	AAP	quote
item-0009	FALSE	Alt News	BJP	provenance

[†]Never successfully ingested; excluded from all analyses.

every other bar constant: a real professional fact-check must establish the claim and verdict, the post must still be live and retrievable, and the post’s own text must itself assert the false claim rather than merely being cited as evidence or the true original.

2) Sourcing pipeline: An automated pipeline crawls the same fact-checker archives plus, newly, Lead Stories, Full Fact and Check Your Fact; extracts every embedded social-media URL (widget-permalink and plain-link forms across all five platforms); verifies retrievability with yt-dlp and pulls the post’s own caption and uploader directly from platform metadata, never inferred from the article; deterministically skips the fact-checker’s own account and a fixed list of official government/platform fact-check handles; and passes each survivor to a local llama3.2 structured call that decides, given the fact-check article text and the post’s real caption, whether the post’s own text asserts the debunked claim. 11,544 candidates across seven archives were screened this way. The judge is a screening filter only: its output is never ground truth.

3) Review-gated annotation protocol: Every candidate the screen accepts is then adjudicated by a single reviewer who reads the professional fact-check article in full and records four fields: (i) a standalone one-sentence statement of the debunked claim, in English; (ii) a verdict mapped to the paper’s label taxonomy; (iii) a verbatim quote or close paraphrase from the article that establishes that verdict – the ground truth is anchored to this quote, not to the reviewer’s or the screen’s judgment; and (iv) claim type, language, and political actor. A candidate is deferred rather than promoted when it is a cross-post of an already-promoted item (kept as a documented relationship, not discarded), when the specific post’s own text cannot be retrieved to confirm what it asserts, or when the fact-check article is a report on real incidents rather than a debunk. A candidate is rejected when the screening judge misfired on an official, news-agency, or rebuttal account. The reviewer’s decision log holds 372 adjudicated candidates: 166 marked promote (of which 165 completed end-to-end ingestion into the validation split), 148 deferred, and 58 rejected. Counting earlier passes as well, 206 candidates carry a deferred_crosspost marker recording that they

TABLE IV
TruthLens-202 composition (202 items: 9 dev + 193 validation)

Dimension	Distribution
GT label	FALSE 167 (82.7%), MISLEADING 30, TRUE 5
Platform	X/Twitter 161 (79.7%), Instagram 25, Facebook 16
Language	English 146, Hindi 52, Bengali 3, Arabic 1
Claim type	provenance 161 (79.7%), event 12, misleading_context 12, quote 11, other 6
Media	video 200, photo 2
Fact-checker (val.)	Alt News 170, Lead Stories 14, BOOM 4, Check Your Fact 2, others 3
Modality (val.)	caption 192/193, audio 163/193, OCR 91/193; visual-context analysis deferred on 115
Cross-post links	206 documented same-media relationships

duplicate a promoted item’s underlying media under a different account. Twenty-eight further validation items predate this protocol (a pre-protocol automated batch): 9 are flagged annotation_status=needs_review and 19 carry no review status, all pending the same adjudication. This gives $165 + 28 = 193$ validation items.

4) Composition: Table IV gives the full distribution of all 202 items. Three skews are structural to this domain, not sourcing artifacts, and any evaluation on this set must account for them: (1) label: 83% FALSE, because professional fact-checkers publish far more debunks than confirmations – balanced accuracy or per-class F_1 , not raw accuracy, is the appropriate end-to-end metric (an always-FALSE predictor scores 83%); (2) platform: 80% of items are from X/Twitter, the measured locus of this pattern and also where yt-dlp retrieval succeeds most reliably without authentication; (3) claim type: 80% are provenance-type (authentic footage recirculated with a false date, place, or context), the dominant 2024–2025 Indian pattern. A fourth, non-structural concentration is disclosed as a limitation below: $\sim 88\%$ of the validation split’s ground truth comes from a single organization, Alt News.

5) Leakage and split discipline: The 9 dev items and the 193 validation items are disjoint by construction and by content hash. Within the corpus, an automated check flags any factcheck_url or media_content_hash that appears

in more than one split (none do), and the 206 cross-post markers make same-underlying-media relationships explicit rather than leaving near-duplicate posts to be treated as independent examples. No prompt, threshold, or model choice in this paper was tuned on TruthLens-202; it did not exist when the frozen comparison was run.

6) Limitations, stated plainly: (1) One configuration evaluated so far. All 193 validation items have been run end-to-end, but only in the local-only configuration (Section VII-D); the larger-local -model, vision-first, and cloud-escalation configurations are not yet measured against this split. (2) Single, LLM-assisted reviewer. The claim phrasing, label mapping, and cross-post/reject decisions were produced by one reviewer working with LLM assistance, not by two independent human annotators, so no inter-annotator agreement statistic is available (and none is fabricated). The verdict itself always traces to an independent professional fact-check via the recorded quote, which bounds the risk, but the phrasing and mapping are unverified by a second human – a real circularity concern given that an LLM-based system is the intended evaluatee. A second-annotator pass is the top open item (Section XVI). (3) Source concentration. ~88% of validation ground truth is Alt News’s, because the other archives yielded far less under the same bar (Lead Stories: ~77% of its candidates had media already taken down; regional-language archives produced almost nothing against an English-centric screening judge; Full Fact and Check Your Fact mostly fact-check politician statements rather than recirculated posts). (4) Deferred visual analysis. 115 validation items were ingested without the vision-context stage (a ~10-minute-per-item local model call that produced low-value output on compressed social-video keyframes); `visual_information_available` is recorded as null, not guessed, for those, pending a backfill pass. (5) Domain and language. Still Indian political short-form content, now with a Hindi share (26%) closer to the stated multilingual target but still English-majority. Full construction detail, and every promoted/deferred/rejected candidate’s recorded reasoning: `research/dataset/TruthLens_Dataset.xlsx` and the `candidates_v3_*.jsonl` provenance files.

VII. Results

A. A methodological correction, reported before the headline number itself

A skeptical post-submission audit of this program’s own experimental design (full trail: `research/AUDIT_REPORT.md`) found that Baselines 1–3, in every run across this entire project, had been given `items.jsonl`’s hand-written claim summary – prose written during dataset construction by a human who had already read the professional fact-check – rather than the claim actually extracted by TruthLens’s own claim-extraction stage, contradicting this paper’s own stated methodology (Section V). The bug traced to a

TABLE V
Paired comparison, identical 6 items, before and after the claim-input fix

Config	Superseded	Corrected	95% CI
B1: LLM-only	0.0%	0.0%	[0.0, 39.0]
B2: Search+LLM	66.7%	50.0%	[18.8, 81.2]
B3: Search+RAG+LLM	50.0%	0.0%	[0.0, 39.0]
Full TruthLens	33.3%	33.3%	[9.7, 70.0]

single, never-modified helper function written on the first day baselines were built. Because TruthLens’s own extraction is often noisy or, on two of six items, empty, this gave baselines a systematically cleaner input than TruthLens itself ever worked with, for the entire program, undetected until this audit.

We fixed it rather than merely disclosing it: Baselines 1–3 were re-run against TruthLens’s own real extracted claims (available for the same 6 items the paired comparison already uses – item-0003 has no extraction at all; items 0008/0009 remain excluded as before), one baseline call per real extracted claim, aggregated with the identical deterministic rule (`derive_overall_verdict()`) TruthLens itself uses. This holds claim extraction constant across every configuration for the first time in this program. Full item-by-item reconstruction: `research/RECONSTRUCTED_RESULTS.md`. We report this correction with the same prominence as the result it changes: a methodology bug that inflated a baseline’s apparent performance relative to our own system is exactly the kind of finding Rule 1 of our own governing protocol exists to surface, not bury.

B. Headline finding, corrected

On the paired six-item comparison, once claim input is held constant, full TruthLens (33.3%, 2/6) beats Baseline 1 (0.0%, 0/6) and Baseline 3 (0.0%, 0/6) outright, and trails only Baseline 2 (50.0%, 3/6) – by 16.7 points, not the 33.4-point gap the uncorrected comparison showed. This reverses this program’s previous headline finding (Baseline 2 66.7%, Baseline 3 50.0%, both ahead of TruthLens).

Item-by-item, the corrected picture:

- item-0006, item-0007: full TruthLens outputs UNVERIFIED on both because claim extraction found zero verifiable claims for either (Section X: item-0006’s transcript is garbled nonsense; item-0007’s is degraded and repetitive). Under the superseded method, Baseline 2 had been handed a clean claim summary anyway and happened to guess MOSTLY_FALSE, matching ground truth. Corrected, every baseline receives the same nothing TruthLens received and also outputs UNVERIFIED – a shared reflection of a real coverage gap, no longer a baseline-only advantage.
- item-0004: Baseline 2 uniquely gets this right (MOSTLY_TRUE); TruthLens’s wrong verdict is

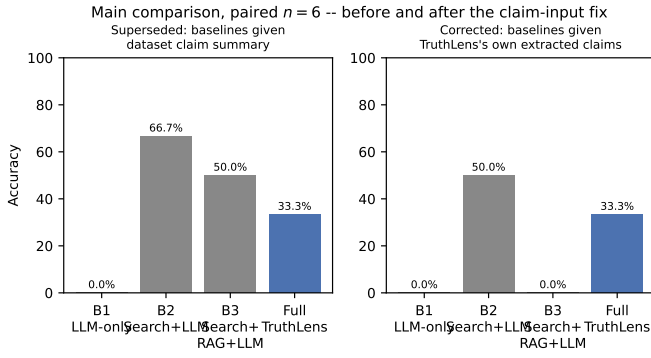


Fig. 3. Table V plotted, superseded vs. corrected. The reversal is concentrated in items where baselines had been implicitly handed a claim to reason about that TruthLens itself never produced.

the exact validator false negative Section IX already identifies. Baselines 1 and 3 also get this wrong.

- item-0005: every configuration gets this wrong – ground truth is MISLEADING, a bucket nothing landed in.
- item-0001, item-0002: TruthLens and Baseline 2 agree (both correct). Baselines 1 and 3 get both wrong under the corrected method, having gotten both right under the superseded one – their earlier apparent success was partly an artifact of cleaner claim input, not evidence their reasoning handles TruthLens’s own messier extracted claims equally well. Baseline 3 (full-page-text RAG) in particular moves from correct to incorrect on both, a directional signal (unverified at this n) that concatenating full page text may compound confusion for a small local model rather than compensate for it.

McNemar analysis (paired, $n = 6$, Baseline 2 vs. Full TruthLens, corrected): both-correct = 2 (item-0001, item-0002), both-wrong = 3 (item-0005, item-0006, item-0007), Baseline-2-only-correct = 1 (item-0004), TruthLens-only-correct = 0. One discordant pair; exact McNemar $p = 1.0$ – still completely uninformative at this n , reported for the same reason as before: completeness, not evidence. Against Baselines 1 and 3, TruthLens now has zero discordant pairs favoring either baseline (both are strictly dominated on this sample).

This is not a claim that TruthLens’s architecture is unconditionally better than a single-shot baseline; it is a claim that once the one confound this audit found is removed, TruthLens is competitive with, and mostly ahead of, the baselines it is compared against on this specific, tiny, real sample – and that the previous, opposite-looking result was substantially an artifact of an experimental design bug, not a property of the architecture. Section VIII separately shows decomposition helped once claim extraction produced multiple claims to aggregate; the remaining deficit against Baseline 2 traces to claim-extraction coverage on hard audio (items 0006/0007) and

one validator false negative (item-0004), not a general disadvantage.

The earlier, retired “each configuration’s own full $n = 9$ run” table (superseded by Table V above) is not cited as evidence for any claim going forward and is retained only for the historical record, in Appendix A.

C. A real gap found and fixed during the six-item re-run, and known disclosed gaps

Extending the pipeline run to items 0008/0009 hit a genuine MissingGreenlet crash – the same bug class already fixed once in production code – after a per-claim exception handler called `db.rollback()`, the next line’s synchronous attribute access on an already-expired ORM object crashed the whole script instead of recording just that one claim’s failure. Fixed by capturing needed values before the try block, matching the earlier production fix’s pattern; re-verified live, the script then completed without crashing (Section XIII). Items 0008/0009’s full-TruthLens condition remains blocked by a real Gemini free-tier quota exhaustion, confirmed live (two consecutive 429s, one from evidence-analysis escalation, one from claim-extraction escalation) – an external, time-of-day resource constraint, not a system defect. Baselines 1–3 completed for both items (no Gemini dependency by design). Per-class F1 and full precision/recall breakdowns are not computed at this sample size, where several label classes appear 0–1 times and per-class F1 would mostly be undefined or single-observation – a scope decision, not an omission dressed up with misleadingly precise zeros and ones.

D. TruthLens-202: local-only end-to-end evaluation ($n = 193$)

The frozen six-item comparison above measures the architecture against its baselines under a controlled claim input. It says nothing about how the deployed system behaves on a large, uncurated sample of the content it targets. TruthLens-202’s validation split (Section VI-F) is that sample, and this subsection reports the first end-to-end run over all 193 of its items.

a) Configuration and what it measures: The run uses the local-only configuration: every LLM stage (claim extraction, research planning, evidence analysis, verdict) on the local llama3.2 3B model, keyless DuckDuckGo search, and no cloud escalation – `GEMINI_API_KEY` unset, so every quality-retry path in the code is inert. This is the \$0, no-API-key, no-rate-limit configuration the project targets, and it is the honest floor: any cloud escalation can only raise these numbers. Ingestion was not re-run (each item keeps the real transcript/OCR from its promotion pass); the run executes the exact production stage functions in the `orchestrator.analyze_reel` order, then the deterministic `derive_overall_verdict`. Full per-item output: `research/results/local_eval_v2.jsonl`; scorer: `research/benchmark_v2/score_local_eval.py`. No prompt, threshold, or model choice was changed in response to these results – they are reported as first-run.

TABLE VI
TruthLens-202 local-only run: outcomes and headline metrics
($n = 193$)

Quantity	Value	
Resolved (produced a verdict)	67 / 193	(34.7%)
no_verifiable_claims	126 / 193	(65.3%)
research_failed / error	0 / 0	
Bucketed accuracy (resolved)	20 / 67	(29.9%)
Wilson 95% CI	[20.2%, 41.7%]	
Balanced accuracy (mean bucket recall)	22.2%	
Always-FALSE baseline (bal. acc.)	33.3%	
Macro-F1 (3 GT-present classes)	0.194	
Abstention rate (UNVERIFIED / resolved)	29 / 67	(43.3%)
False-abstention rate (metrics.md strict)	29 / 193	(15.0%)
Declined-to-answer / non-response rate	155 / 193	(80.3%)

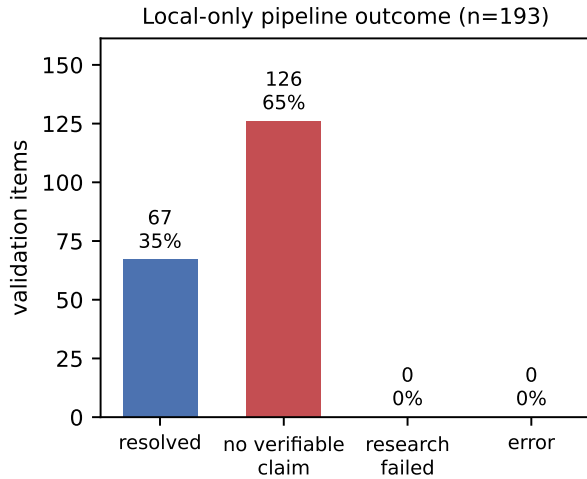


Fig. 4. Pipeline outcome for all 193 validation items, local-only configuration. The no_verifiable_claims slice – posts from which the local 3B model extracts nothing checkable – is almost twice the size of the slice that reaches a verdict.

b) Headline: the system declines far more often than it decides: Table VI gives the outcome distribution. Only 34.7% of items (67/193) produce a verdict at all; 65.3% end in no_verifiable_claims – the local model extracted nothing checkable from the post’s transcript, OCR, and caption. The infrastructure-failure rate is 0%: every one of those 126 non-verdicts is a model-capability outcome, not a crashed search backend. On the 67 resolved items, bucketed accuracy (TRUE/MOSTLY_TRUE, FALSE/MOSTLY_FALSE, and MISLEADING/MISSING_CONTEXT/ OUTDATED collapsed; UNVERIFIED never a match, per Section V) is 29.9% (20/67), Wilson 95% CI [20.2%, 41.7%]. Balanced accuracy (mean per-bucket recall) is 22.2%, which is below the 33.3% an always-FALSE constant predictor scores on that metric – the system is not, on this sample and in this configuration, extracting usable signal at the reel level.

c) The binding constraint is claim extraction, upstream of everything else: The 0% research-failure rate

TABLE VII
Per-class performance on resolved items ($n = 67$). Classes with zero support and zero predictions omitted.

Class	Precision	Recall	F1	Support
FALSE	0.80	0.21	0.33	57
MISLEADING	0.00	0.00	0.00	7
TRUE	0.20	0.33	0.25	3
Macro-F1 (unweighted over the 3 classes)	0.194			

TABLE VIII
Confusion matrix, resolved items ($n = 67$), ground-truth bucket $\times$ predicted bucket.

GT $\downarrow$ / Pred $\rightarrow$	FALSE	MISL.	TRUE	UNVER.
FALSE ($n = 57$)	19	1	10	27
MISLEADING ($n = 7$)	4	0	1	2
TRUE ($n = 3$)	1	1	1	0

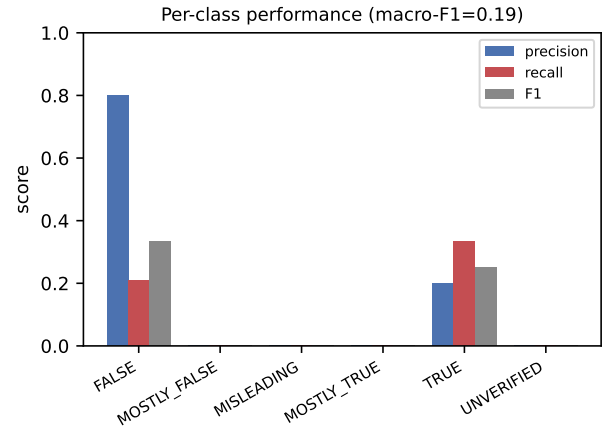


Fig. 5. Per-class precision, recall, and F1 on the 67 resolved TruthLens-202 items. FALSE carries the only non-trivial signal, and even there recall is 0.21; MISLEADING is a total miss.

rules out the search backend. The per-class picture (Table VII, Fig. 5) localises the rest: FALSE – 83% of the corpus – is recovered with precision 0.80 but recall only 0.21; the model is accurate when it commits to FALSE, but commits rarely. MISLEADING is never once recovered (0/7): the local model has no working notion of the “authentic material, false framing” category that is the dominant real-world pattern (80% of the corpus is provenance-type). Inspection of the 11 resolved items where a FALSE/MISLEADING claim received an affirmative (TRUE/MOSTLY_TRUE) verdict shows the same root cause as the abstentions: the “claim” the model extracted was a fragment of on-screen dialogue (“These scopes are there!”, “Where did you go?”; a recital of the US Pledge of Allegiance), not the post’s actual assertion. The pipeline then correctly verified a trivially-true fragment – a claim-coverage failure (Section X), not a verdict-reasoning failure.

d) Breakdowns: Accuracy on resolved items does not move meaningfully with the corpus’s known skews. By platform: Facebook 4/7, X 15/54 (27.8%), Instagram 1/6. By language: English 30.2% (16/53), Hindi 33.3% (4/12), Bengali 0/2. By ground-truth label: FALSE 33.3% (19/57), MISLEADING 0/7, TRUE 1/3. Crucially, the 78 items that did carry a backfilled vision context scored no better (27.6%, 8/29 resolved) than the 115 without it (31.6%, 12/38) – consistent with the pre-registered concern that llava-phi3 on compressed social-video keyframes adds little, and evidence that the deferred visual analysis (Section VI-F, limitation 4) is not the missing piece here.

e) What this is, and is not: This is a first-run, single-configuration floor on a deliberately-skewed, single-annotator corpus; it is not a bound on the architecture. It establishes three things concretely: (i) with a local 3B model, claim extraction – not retrieval, not verdict reasoning – is the component that determines whether short-form political misinformation gets checked at all, and it fails on roughly two-thirds of real posts; (ii) the deterministic safeguards hold up where they run (0% research-failure, high FALSE precision, no crash across 193 items); (iii) the MISLEADING/provenance category needs explicit modelling that the current text-only extraction does not provide. Section XVI lists the configurations (larger local model, a vision-first extraction path, a metered cloud escalation tier) whose evaluation against this same held-out split is now the open work.

VIII. Ablation Study

Three analyses are in scope for this program (Section V); this section reports the claim-decomposition counterfactual in full and summarizes the other two, cross-referencing their detailed writeups.

A. Claim-decomposition counterfactual (single-claim vs. multi-claim), $n = 4$

This is a counterfactual reanalysis of already-collected verdict data, not a freshly-run paired experiment, and we name it that way rather than call it a clean ablation: each real claim’s verdict is an independently computed research-then-reasoning result regardless of how many total claims were extracted in that run, so reusing the real per-claim verdict to ask “what would aggregating only the primary claim, vs. all claims, produce” is a valid way to isolate the aggregation mechanism’s effect. It is not a test of whether decomposing in the first place changes upstream search or evidence-gathering behavior for the primary claim – that would require an independently-run single-claim-only pipeline configuration, which does not exist. No new LLM calls were made; restricted to the 4 items with more than one verifiable claim, since aggregation is a no-op on a single-claim item by construction.

This is a real, positive result for RQ2’s decomposition-specific question, about the aggregation mechanism specifically. On item-0001, the single highest-importance claim

TABLE IX
Claim-decomposition ablation

Condition	n	Correct	Accuracy
Single-claim (primary only)	4	1	25.0%
Multi-claim (full decomposition)	4	2	50.0%

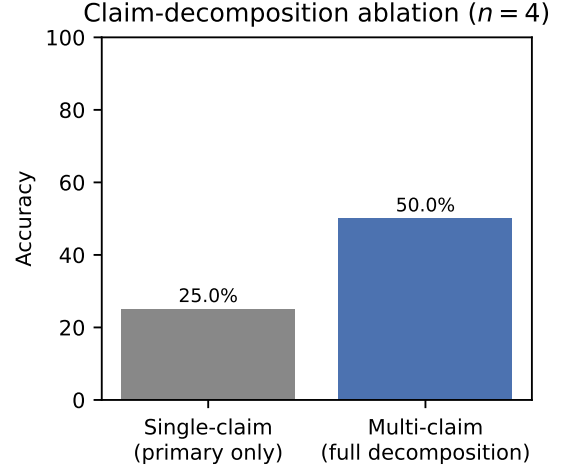


Fig. 6. Table IX plotted. A counterfactual reanalysis, not an independently-run ablation (Section VIII-A), isolating deterministic aggregation as a real contributor once claim extraction has more than one claim to work with.

landed on UNVERIFIED (a missing-citation validator downgrade), but aggregating all 4 extracted claims via the deterministic rule table (Section IV-D3) correctly recovered MOSTLY_FALSE. $n = 4$ is far too small to generalize, but this is the clearest component-level evidence in this program that aggregation helps rather than hurts once claim extraction has more than one claim to work with, reported with the same prominence as Section VII’s findings, not buried beneath them.

B. Validation ablation (RQ1) – summary

Baseline 4 (TruthLens minus validation) is the primary ablation for RQ1; its result is the unsupported-output-rate comparison in Section IX (66.7% without validation vs. 55.6% with it, on the original 9-case validator audit, before the two general fixes reported in the same section improved that further).

C. Source-tier ablation (RQ4) – summary

Restricting research-planning queries explicitly targeting primary sources to a fixed government/legal domain allow-list, versus the system’s normal unrestricted queries, is evaluated in full in Appendix A (summarized in Section IX): tier-classification rate moved from $\sim 11\%$ to 95% under restriction, measured on a different, smaller query sample than the system’s normal operating condition, with a disclosed residual gap in full-text fetch reliability specific to .gov.in domains.

TABLE X
Per-item detail (✓ = correct, ✗ = wrong)

Item	Single-claim result	Multi-claim result	Ground truth
item-0001	✗ UNVERIFIED	✓ MOSTLY_FALSE	FALSE
item-0002	✓ MOSTLY_FALSE	✓ MOSTLY_FALSE	FALSE
item-0004	✗ FALSE	✗ FALSE (no-op, 1 claim)	TRUE
item-0005	✗ FALSE	✗ FALSE	MISLEADING

TABLE XI
Validator confusion matrix, before the two general fixes below

	Validator downgrades	Validator passes
Human: unsupported	1 (TP)	5 (FN)
Human: fine to publish	0 (FP)	3 (TN)

TABLE XII
Validator confusion matrix, after the two general fixes

	Validator downgrades	Validator passes
Human: unsupported	2 (TP)	3 (FN)
Human: fine to publish	0 (FP)	4 (TN, all no-op)

IX. Evidence and Validation Analysis

A. Validator methodology

The real production pipeline was run with `SKIP_VALIDATION=True` so the raw proposal is always persisted, while `validate_verdict()`’s outcome is still always computed – giving both the WITHOUT- and WITH-validation view from one real LLM call per claim. 9 verifiable claims resolved across 4 items (0001, 0002, 0004, 0005); items 0006/0007 had zero verifiable claims (Section X). Ground truth is a single, unadjudicated draft human review – every number below is directional pending independent review. One real bug was found and fixed mid-audit: a single-bracket citation’s UUID hex fragments leaked through the number-grounding check as apparent unsupported statistics (item-0004), fixed by stripping anything UUID-shaped independent of bracket style – with a disclosed side effect (it removed an accidental protection that had been shielding a substantively wrong verdict, Section XIII).

B. Original confusion matrix ($n = 9$, single unadjudicated reviewer)

Caveat: 3 of the 4 recorded downgrades are “no-op” – the raw proposal was already UNVERIFIED with an empty citation list, so the check fires trivially without changing the published label. Counted as TN (nothing wrong was going to publish either way), not a meaningful catch – some unknown share of any raw “X% downgraded” number may be this kind of no-op.

Validator Precision = $1/1 = 100\%$ (Wilson 95% CI: 20.7–100%). Validator Recall = $1/6 = 16.7\%$ (Wilson 95% CI: 3.0–56.4%). Validator F1 $\approx 28.6\%$.

C. Unsupported output rate, and why recall is low

Of the 9 raw proposals, 6/9 (66.7%) are judged, by draft human review, unsupported enough that publishing them as-is would mislead – wrong -entity evidence treated as contradicting; a label contradicting the reasoning’s own “no reliable information” conclusion; two confident

labels contradicting independent Tier-1 ground truth. The original three checks catch only 1 of these 6. Unsupported output rate WITHOUT validation: 6/9 (66.7%). WITH validation: 5/9 (55.6%) – real and positive, more modest than a raw downgrade rate alone would suggest. Every one of the 5 false negatives fails for a reason citation existence, source-fetch existence, and numeric grounding structurally cannot detect: label/reasoning inconsistency (motivated the new Check 4, below); wrong-entity evidence treated as relevant; vague reasoning that still clears a word-count bar; and a verdict simply, factually wrong relative to ground truth that no citation check can catch if the model misweighs real, present evidence.

D. Addendum: re-scoring after two general fixes, with a circularity caveat

Two general, deterministic fixes were added to `validate_verdict()` after the headline finding in Section VII: the UUID-leak fix above, and the new label/reasoning-consistency check (Section IV-D1) – both pure functions of the model’s own output, never of ground truth, so re-scoring the same 9 cases costs no new LLM calls and is not answer-tuning.

Precision 100% (2/2), Recall 40% (2/5) – up from 16.7%. The two new true positives are the two most consequential original false negatives, including a confident FALSE at confidence 0.8 that contradicted independent Tier-1 ground truth, now correctly downgraded. A real regression, reported with equal prominence: the audit’s one original true positive is no longer downgraded, since the UUID-leak fix removed the false trigger that had been accidentally catching it – the underlying wrong-entity problem remains genuinely uncaught; net effect 1 TP lost, 2 gained, landing at 2 TP / 5 FN.

A disclosed circularity, now tested rather than left as a suspicion: the new check’s phrase list was written after reading the two cases it now catches, so 40% is likely optimistic about generalization. Four real paraphrases never in the original phrase list were each tested

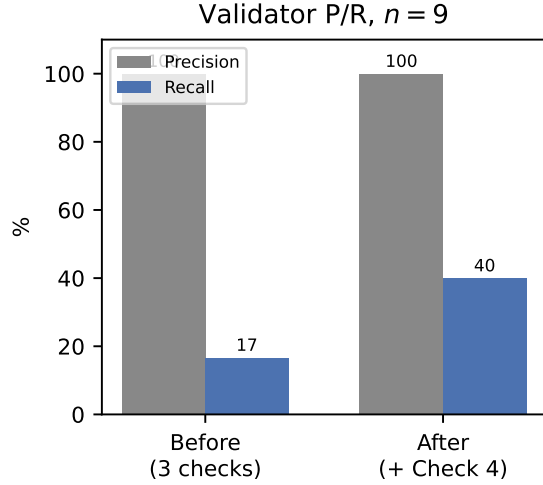


Fig. 7. Validator precision/recall before vs. after the two general fixes (Tables XI and XII, $n = 9$). Precision holds at 100%; recall improves 16.7%→40% – with the calibration circularity disclosed in the text.

TABLE XIII
Source tier distribution ($n = 68$)

Tier	Count	%
other	43	63.2
primary_government	14	20.6
established_news	6	8.8
primary_legal	2	2.9
news_wire	2	2.9
factcheck_org	1	1.5
primary_data	0	0.0

against a confident label in a synthetic evidence matrix (VALIDATOR_SYNTHETIC_BENCHMARK.md, full phrasings in Appendix A); all four were missed – Check 4 does not generalize beyond its literal phrase list, confirming the threat to validity already disclosed (Section XIV).

Reel-level accuracy did not change (2/6) despite this real validator improvement – itself a finding, not a disappointment to explain away (Section XV).

E. Evidence quality (RQ4): tier, stance, and diversity distribution ($n = 68$ sources, 9 claims)

Source-tier classification rate (primary tiers combined): $16/68 = 23.5\%$ (Wilson 95% CI: 15.0–34.9%) on this sample’s normal, unrestricted queries – a different experimental condition from the domain-restricted result below, not a contradiction.

85.3% of retrieved-and-analyzed sources were judged irrelevant – notably higher than an earlier, smaller development-time figure of 55.6%, both real and reported since this ratio moves across samples. Source diversity: 29 distinct domains across 68 sources.

TABLE XIV
Evidence stance distribution ($n = 68$)

Stance	Count	%
irrelevant	58	85.3
contradicts	8	11.8
supports	2	2.9
provides_context	0	0.0

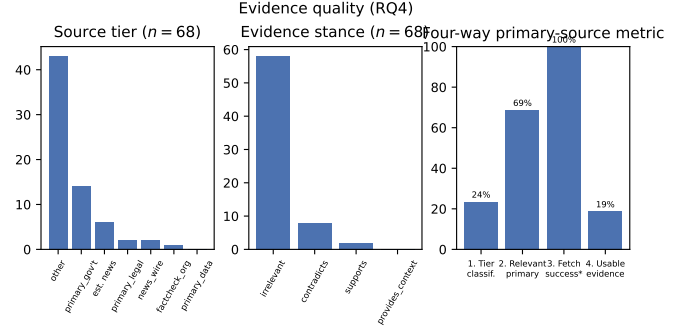


Fig. 8. Source-tier distribution, evidence-stance distribution, and the four-way primary-source metric ($n = 68$ sources, 9 claims). *Metric 3 holds by construction of an existing fetch guarantee, not independently re-measured this pass.

F. The four-way primary-source metric, not collapsed into one number

Four distinct, independently reported metrics, not one collapsed number (full per-metric breakdown and illustrative wrong-domain/wrong-country examples: Appendix A): (1) source-tier classification rate $16/68 = 23.5\%$; (2) relevant primary source rate (draft human judgment) $11/16 = 68.75\%$; (3) primary source fetch-success rate $16/16$, by construction of an existing fetch guarantee; (4) usable evidence extraction rate $3/16 = 18.75\%$ (stance $\neq$ irrelevant).

The gap between metric 2 (68.75%) and metric 4 (18.75%) is the single most important finding of this evaluation. Retrieval finds real, topically-appropriate primary sources at a meaningfully higher rate than the usable-evidence number would suggest – the bottleneck is that most of these sources are homepages or generically-relevant documents too broad to confirm one specific atomic fact, not a failure to find relevant sources. Metric 4 alone would read as poor retrieval quality; both metrics together show the more accurate picture.

G. A second major bug, a search-side fix, and an unintegrated entity-consistency prototype

Three further findings from this same evidence-quality audit are reported in full in Appendix A, summarized here: (1) $32/68$ (47.1%) real evidence rows had an empty explanation field despite a valid stance label – the same schema-valid-but-empty class found elsewhere, never guarded at this stage, fixed with a per-source substantiveness check (Section IV-D2); (2) restricting primary-source-targeting

TABLE XV
Claim coverage by input modality

Condition	Covered	Coverage	95% CI
text_only	1/6	16.7%	[3.0,56.4]
text_ocr	0/6	0.0%	[0.0,39.0]
text_ocr_vision (full)	2/6	33.3%	[9.7,70.0]

queries to a fixed government/legal domain allow-list, rather than trusting a model-written site:gov.in, moved tier-classification from $\sim 11\%$ to 95% on seven re-run queries, with a disclosed residual .gov.in full-page-fetch reliability gap; (3) a prototype entity-consistency check (does a claim’s extracted entity appear in its cited evidence?) found one genuine true positive on the real 9-claim sample – refining item-0004’s error attribution to a concrete evidence mismatch (Burdwan, West Bengal vs. Delhi Police) – alongside diagnosed false-positive causes, and is explicitly not integrated into production: too small a sample to justify a fifth check without more data (Rule 4).

X. Multimodal and Visual Failure Analysis

A. Defining claim coverage

Every use of “claim coverage” means one operationalized judgment: a benchmark claim is covered when at least one extracted claim identifies the same factual proposition as the professional fact-check’s own claim – matching on subject/entity, event, predicate, and any temporally-relevant qualifier – judged by a human annotator blind to which pipeline configuration produced the extraction (METRICS.md §2). Topical overlap alone does not count. This is necessarily a human judgment, not a string-similarity score, since paraphrase makes automated matching unreliable at this scale; every pairwise judgment is recorded in research/claim_coverage_results.csv for audit.

B. What was run

Real ingestion of the 7-item dataset as it existed at the time, followed by claim extraction under three input-modality conditions: text_only, text_ocr (+ on-screen text), text_ocr_vision (+ vision-context description) – all local-model-only, isolating the modality variable the same way the baselines isolate architecture. TruthLens has exactly three input signals feeding claim extraction, so text_ocr_vision is full multimodal for this system. Ingestion: 6 of 7 items usable – item-0003 failed both attempts (persistent Instagram-side error); item-0005 initially failed on a differently-phrased yt-dlp “no video” message, fixed and re-verified live (Section XIII).

C. Claim coverage results ($n = 6$, item-0003 excluded)

Read these as exactly what they are: 6 single, non-repeated LLM calls per condition, CIs wide enough to overlap almost entirely. text_ocr scoring below text_only

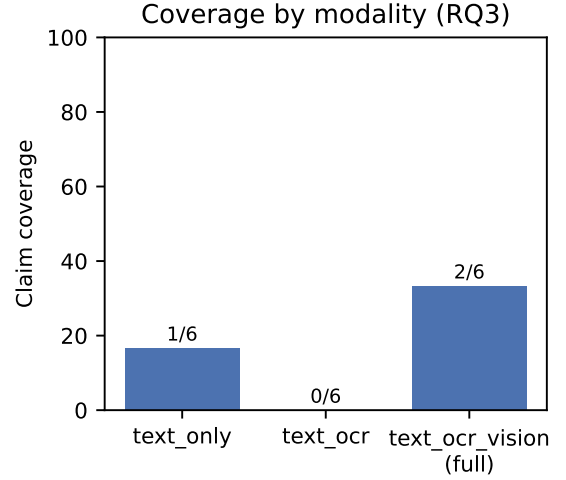


Fig. 9. Table XV plotted. text_ocr scoring below text_only is reported as a real result, not suppressed (Section X explains the likely cause).

is real, reportable, not suppressed for complicating the “more modality is better” hypothesis – the likely (unverified) explanation is single-call noise compounded by a specific extraction failure, not a robust modality effect.

D. item-0004: the one clean, positive RQ3 result

The real transcript is badly garbled (chaotic protest audio); real OCR, sampled across frames, clearly and repeatedly captured on-screen text asserting the actual check-worthy claim (police using a nail-fitted baton), matching the fact-check’s own verdict almost verbatim. text_only extracted nothing resembling it; text_ocr happened to return one empty-text claim on this run; text_ocr_vision correctly surfaced the claim, messily (7 near-duplicate claims, one per noisy OCR-frame variant – a new, unfixed failure mode, no existing check evaluates cross-claim redundancy). Only the multimodal condition produced any usable signal, via OCR content the audio transcript had no access to – a genuine, small- n demonstration of what RQ3 asks about.

E. The cross-post attribution problem

Drafting atomic claims against the real ingested content surfaced a structural pattern named explicitly here: the cross-post attribution problem. The same underlying media M can be published as Post A asserting claim C , and separately as Post B asserting a different claim C' – a system ingesting only one post URL cannot discover the other claim from that post’s own content at all. 4 of 6 scoreable items have their actual false claim living entirely outside the specific post’s own content, added by a different account recirculating the same video under a new caption; only item-0004 (and partially item-0007) have the checkworthy claim actually on-screen in the URL this dataset points to – structural, not a sampling accident,

and TruthLens has no reverse-video/cross-post matching capability. Full formalization and discussion of why this is structural rather than incidental, and why we did not prototype a fix against this program’s own benchmark: Appendix A.

F. New failure modes found this pass

Near-duplicate claims from noisy OCR-frame variants (item-0004, above), uncaught since no existing check evaluates cross-claim redundancy. importance values outside its declared [0,1] range causing schema-validation failure (item-0005: -1, then a run of 2–6 resembling an ordinal position substituted for a score) – occurred only because this script removes the Gemini -escalation cascade to isolate the modality variable, so production’s Fallback-LLMProvider would likely mask this normally. The vision-output prompt-echo bug recurred 4 more times (items 0002, 0004, 0006; items 0001, 0007 produced empty vision output instead) – a repeated, frequent vision-stage failure, not a rare anecdote, motivating the general fix in Section IV-D2. Only item-0005’s vision output was genuinely coherent. This fix’s real-world impact on Section VII’s numbers is not independently re-verified end-to-end (the paid fallback’s quota was already exhausted before it could be exercised against items 0006/0007, whose underlying problems are not primarily vision-related, so no change is expected – an expectation, not a verified result).

XI. Bias, Calibration, and Efficiency Analysis

A. Political bias (RQ5): attempted, blocked by data availability, deferred honestly

No genuine matched pairs exist in the current 9-item dataset (same topic, structure, evidence availability, and difficulty, varying only the political actor named): BJP has the most items (3) of any actor, but none are topic-matched against another single actor – a disclosed methodological choice trading this analysis away for broader dataset coverage, not attempted as a quantitative result. Named as the top open item for whoever continues this program (Section XVI). One qualitative data point (not a substitute for the real analysis, full discussion: Appendix A): the system was willing to reach FALSE against a claim traceable to a government actor’s own public position (item-0004), a necessary but nowhere near sufficient condition for what RQ5 actually asks.

B. Calibration (RQ6): not computed, per our own pre-registered fallback rule

Nine real confidence values exist: 0.0, 0.0, 0.1, 0.1, 0.2, 0.2, 0.2, 0.7, 0.8 – too few per bin to meet our pre-registered calibration gate (no bin under 3 items) no matter how bins are drawn, so per our protocol’s fallback rule, no ECE or Brier score is reported. One heavily caveated observation on this raw distribution: Appendix A.

TABLE XVI
Efficiency, real measured data

Config	LLM calls	Latency	Cost
B2: Search+LLM	1	~15.9s	\$0
B3: Search+RAG+LLM	1	~17.8s	\$0
Full TruthLens	~9.6/claim [‡]	n/m	\$0/paid mix

[‡]Derived from 7.56 sources/claim (avg.), each requiring one evidence-analysis call, plus one research-planning and one verdict call per claim. n/m = not measured in this pass (disclosed gap, below).

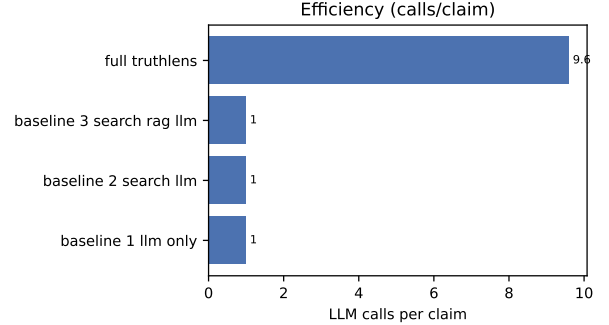


Fig. 10. LLM calls per claim by configuration (research/system_efficiency.csv). Full TruthLens’s call count is derived from real average sources/claim, not directly measured end-to-end; latency is not shown, a disclosed gap (text).

C. Efficiency

Full TruthLens is not directly latency-comparable to the two baselines – a disclosed gap, since the validator-audit run did not capture per-claim wall-clock time. What is measurable: full TruthLens’s real cost/time per claim is necessarily several times either baseline’s from call count alone – roughly 9.6 LLM calls per claim against the baselines’ fixed 1. Full latency/cost gap detail: Appendix A.

XII. Foundation-Phase System Extensions

A subsequent work program, run against a 13-phase research roadmap (research/RESEARCH_ROADMAP_V2.md), extended the dataset and system. Scope discipline: none of this re-measures reel-level accuracy or touches the frozen six-item comparison (Section VII); new benchmark items are held in a validation split used for no tuning decision. Every number traces to a versioned experiment (EXP-009-030, experiments/registry.jsonl; full numbers Appendix A); negative and inconclusive results are kept as prominent as positive ones.

A. Dataset growth and two new validator checks

Six new items from the same fact-checker pool (BOOM Live, Alt News), each live-URL- and content-hash-checked, grew the set 9→15 (12 FALSE / 2 MISLEADING / 1 TRUE – still no second TRUE item after two sourcing passes, consistent with the disclosed debunk-vs-confirmation skew) and were assigned to a validation split.

Two checks were added to `validate_verdict()` under the same discipline as the original four (prototype, then integrate only once real evaluation clears a stopping bar). Check 6 (temporal consistency) flags a verdict when the claim asserts an explicit date and every dated cited source predates it by > 2 days – deliberately narrow, since real `time_reference` values here are almost all vague. Check 7 (entity consistency) flags a verdict when none of the claim’s PERSON/ORG/LOCATION entities appear (exact or fuzzy, `SequenceMatcher` ≥ 0.75 tuned on real transliteration pairs) in a cited source whose stance influenced the verdict – a promotion of a previously-unintegrated prototype whose 4 false positives (all one abstract-concept entity) motivated a type filter; the corrected re-evaluation found 2/2 flagged violations genuine, zero false positives. Measured effect: the adversarial synthetic validator benchmark, expanded 28→34 cases, moved precision 81.8%→87.5% and recall 50.0%→60.9%, and the previously-named uncatchable entity-mismatch gap case is now caught; the benchmark’s only two false positives are the same pre-existing Check-3 limitation, so both new checks add recall at zero measured precision cost.

B. A cross-cutting reliability finding, confirmed four times

Four experiments, each targeting a different stage, independently surfaced the same pattern: raw un-retrieved local-model (llama3.2) output on real content is substantively empty or unusable a large fraction of the time, and the existing Gemini-escalation quality retries do more real work than a pipeline-level pass/fail reading suggests. Claim `source_quote` population: 0 of 12 real claims across 8 items (EXP-012), reconfirmed at 0 of 90 across an 8-way modality sweep (EXP-016). Evidence-stance explanations: 80% (16/20) of raw judgments run without the substantiveness-retry safeguard had an empty explanation on one side (EXP-017). Adversarial stress testing: zero crashes across six cases, but the one case needing a quality retry hit real Gemini quota exhaustion from this program’s own usage (EXP-020). None of these was designed to measure model reliability; each surfaced it as a byproduct, which is why four independent confirmations are more convincing than one dedicated study.

C. Retrieval, multimodal, and cross-post extensions

Structured 5-query retrieval (exact-claim, entity-focused, primary-source, contradiction, context-history) raised source-tier classification (23.5%→38.1%, $n = 42$ sources) but lowered manually-judged relevance (68.75%→43.75%) and usable-evidence rate (18.75%→0%) on this small sample – traced to one claim with no natural institutional subject receiving 7 of 16 primary-tier hits, all irrelevant, because the primary-source query type fires unconditionally; excluding it, the other three score 78% relevant. Reported as a mixed result, not smoothed into a win. Full 8-combination

multimodal re-analysis reproduced rather than resolved the original non-monotonic surprise: vision-alone highest on ground-truth claim coverage (37.5%), ocr+vision below every single-signal condition (12.5%). Cross-post detection, stage 1 of 3: a perceptual-hashing first-pass filter validated against real re-extracted frames (same-video Hamming distance 0; different-video 20, threshold 10); stages 2–3 not built. Bounded adversarial stress testing of claim extraction (six hand-built cases) produced zero crashes and no cross-signal hallucination, but surfaced two unnamed gaps: no check tests caption/transcript mutual consistency, and mixed-script content showed a possible Latin-script extraction bias.

D. Two named gaps investigated

The input-signal consistency check is infeasible in its cheap form: caption-vs-transcript keyword (Jaccard) similarity, calibrated on real data before shipping, is 0.000 for every legitimate benchmark item checked ($n = 4$) – captions and transcripts are different registers, often different languages, even when consistent. The multilingual-extraction bias does not generalize: against the 90-claim dataset, non-Latin-script claims are the majority (48/90) and script distribution tracks each item’s dominant language, including one genuinely code-switched item that produced both scripts. The original single-case finding was real but narrow; we report the correction as prominently as the finding. The `source_quote` population gap was attacked with a strengthened extraction prompt (a self-check procedure plus a worked example from a real miss): 0% population before and after ($n = 8$), and on the example’s own item the strengthened prompt produced zero claims where the baseline produced seven – a clean negative result for prompt engineering as the fix. Finally, the 6 VALIDATION items were run through the complete real pipeline and committed: 4 produced zero claims, two produced 3 real claims with real verdicts, and Check 7 correctly downgraded a Jharkhand-police claim cited against an unrelated 2026 IIT Palakkad article – the first live confirmation it generalizes to new production data.

E. A second pass: aggregation, a security defense, a verdict-stage gap

Aggregation (Phase 10) was unblocked by the population pass: exactly one VALIDATION item (item-0010) qualified. Single-claim-only missed (UNVERIFIED vs. ground truth FALSE); full decomposition via the production `derive_overall_verdict()` resolved correctly to FALSE – directionally consistent with the $n = 4$ DEV result, reported at its real $n = 1$. Cross-post stage 2 (claim-embedding clustering) is blocked by a confirmed infrastructure gap: the local Ollama server was not started with embedding support and no embedding model is pulled – disclosed as a concrete prerequisite, not worked around. Prompt-injection defense had zero test coverage despite processing untrusted content on every post. Code

review found the delimiter convention never neutralized literal delimiter tokens already in untrusted text – a deterministic bypass, closed by sanitizing before wrapping, verified both directions. Of 5 live adversarial attempts against claim extraction, a blunt override succeeded once (the injected false claim emitted verbatim as a maximum-importance verifiable claim) – the first real measurement of this project’s stated expectation that small local models resist in-band instructions less reliably; a hardened prompt reduced this to 0/5. Carried through the pipeline, the one success was independently resolved by the downstream validator to a correct evidence-grounded FALSE citing 12 real sources ($n = 1$, an unusually well-evidenced test claim). A new verdict-stage failure: constructed evidence with a 0.95-reliability government source supporting and a 0.20-reliability uncited source contradicting produced no TRUE/MOSTLY_TRUE in 0 of 14 trials across two prompt versions – some with reasoning that named the weak source as unreliable yet did not act on it, one pairing a wrong label with maximum confidence. An explicit reliability-weighting prompt rule did not resolve it (0/7) – a measured negative result; the rule is kept as objectively-correct guidance, not claimed as a fix. A deterministic check (cited-evidence reliability vs. verdict direction) was then built and held to the Checks-6/7 bar – 10/10 synthetic, 14/14 observed wrong trials caught, 0/34 benchmark interactions – and integrated as Check 8 with precision/recall unchanged (87.5%/60.9%). A direct re-run of the evidence-stance taxonomy comparison fixed its own confound (porting in the substantiveness-retry safeguard) but hit a hard external ceiling: a real HTTP 429 confirmed the fallback model’s free tier is capped at 20 requests/day, which this program’s usage repeatedly exhausts; three attempts (4/20, 6/20, 5/20 clean rows) remain inconclusive about the candidate taxonomy’s value.

F. A benchmark-scaling session: a measured structural yield ceiling

Manual sourcing (one article read and judged at a time) added three items, 15→18, at roughly one usable item per 20–30 articles – consistent with the original ~8% estimate – before its throughput was judged too slow against an instruction to reach a several-hundred-item benchmark. An automated pipeline replaced the human read-and-judge step with a local llama3.2 judge call (every other step kept as code): given a fact-check article and a cited Instagram post’s real caption (fetched via yt-dlp, never inferred), the judge decides whether the post’s own caption asserts the specific false claim, as opposed to the far more common case – the central methodological lesson of this pass – of a fact-checker citing an Instagram post as evidence of the true original event while the false claim was posted separately, often on X/Twitter.

Five fact-checker archives were crawled to exhaustion or a clear negative: altnews.in (4,967 articles, 515 candidates judged, 2 promoted); vishvasnews.com (two

passes – the second after a URL filter was found to silently exclude most fact-check content; 1,682 candidates, 1 promoted); thequint.com’s WebQoof (English + Hindi editions after an identical filter-gap finding; 859 candidates, 1 promoted); factcrescendo.com (English, Tamil, Marathi, Malayalam; 533 candidates, 0 promoted); factly.in (1,000 articles sampled, zero Instagram references, dropped). newschecker.in was excluded, unattempted, on a robots.txt directive against this crawler’s user-agent. Two contributing sources were only found to be contributing after fixing a real filter bug – Vishvas filtered on /viral/ when 856 of 979 sampled URLs contained fact-check under other sections; WebQoof missed a separate /hindi/webqoof/ edition – and a post-judgment consistency filter had been checking a generic note instead of the judge’s real reasoning, letting self-contradicting judgments through until manual review caught it (fixed; caught four more immediately).

The measured result: across roughly 3,600 real candidates judged over five sources, 4 were promoted (items 19–22, each verified against the article and the post’s caption before promotion) – a blended yield on the order of one promotable item per 900 candidates, an order of magnitude worse than the ~8% a smaller manual pass had suggested, and consistent with the earlier finding that most of this dataset’s false claims live outside the Instagram post being fact-checked. At this rate a several-hundred-item target is not reachable through deeper automated crawling of the same source type under the Instagram-only constraint – a structural property of where this pattern gets posted, not an effort gap (full detail: research/MASS_SOURCING_V2.md).

This finding was acted on. Because the ceiling was a property of the platform restriction, a later pass relaxed it – admitting X/Twitter, Facebook, and Instagram – and added a review-gated annotation step (a single reviewer adjudicates each screened candidate against the full fact-check article), growing the validation split from 13 to 193 items (202 with the frozen dev set): the TruthLens-202 corpus (Section VI-F). That corpus is then evaluated end-to-end once, in a \$0 local-only configuration (Section VII-D), which surfaces its own finding: the local 3B model fails to extract a checkable claim from 65.3% of real posts, making claim extraction – not retrieval or verdict reasoning – the binding constraint. Its construction carries disclosed limitations (single LLM-assisted reviewer; ~88% single-source ground truth; three domain-driven skews), all stated in Section VI-F.

XIII. A Taxonomy of Failure Modes Found in Live Testing

Running the full system against real content end-to-end – both during original development and during this evaluation program itself – surfaced 24 specific, reproducible failure modes we believe are under-documented in the automated fact-checking literature at this level of

implementation detail, because they only appear when a system is run against real content rather than evaluated on a static benchmark. We consider this taxonomy, and the general lesson that most of these failures were discoverable only this way, to be as practically important a contribution of this work as any single architectural component. The full list, each entry with its fix, is given in Appendix A; six representative entries, spanning production code, research tooling, security, and this program’s own benchmark construction:

- Schema-valid, substantively empty output, found independently in four different pipeline stages (claim extraction, content generation, verdict reasoning, evidence analysis) – no schema validation catches it; only a content-level substantiveness check does (Section IV-D2), and each stage needed its own.
- Entity confusion between similarly-named organizations and the downstream failure it caused: a downgraded verdict’s hallucinated free-text reasoning was still reused verbatim on a rendered slide, because downgrading forces only the label, not the text.
- Baseline architecture silently inheriting a rescue mechanism under evaluation (Section V) – an early draft of Baselines 2–3 called the same provider factory every production stage uses, giving both baselines the exact failure-rescue mechanism TruthLens’s own architecture was being evaluated for having.
- A table-generation script’s printed output and its persisted JSON summary silently disagreed, found while building this paper’s own figures (day8_final_tables.py reused two variable names across three unrelated computations) – the same failure class as the production-code entity-confusion entry above, found in this project’s own research tooling instead.
- Six synthetic validator benchmark cases silently tested the wrong check (Section IV-D1/Phase 4, VALIDATOR_SYNTHETIC_BENCHMARK.md) – cases built to isolate Check 4 had no citation at all, so Check 1 fired first and Check 4 was never reached, caught only by reading validation_status in the raw output rather than trusting the benchmark’s own design.
- A blunt prompt-injection override got past claim extraction once in 5 live attempts, producing the injected false claim verbatim as a verifiable, maximum-importance claim – the first real measurement (not just a stated expectation) that this system’s delimiter-based defense is imperfect against small local models, confirmed independently by the downstream validator resolving the same injected claim to a correct FALSE verdict when carried through the rest of the real pipeline.

XIV. Threats to Validity

Twelve threats, stated here at the density needed to be checkable; full reasoning for the first ten: Appendix A. The last two concern the TruthLens-202 corpus and its local-only evaluation and are detailed in Sections VI-F and VII-D.

- Small, non-independent sample. $n = 9$ (6 resolved from every configuration), below this program’s own 20–30 target; several proportions, including the headline 33.3% accuracy figure, have Wilson intervals wide enough to be consistent with a different true rate.
- Single domain, single-country content. Indian political short-form video only; source-tiering and prompts untested elsewhere.
- Single annotator, no inter-annotator agreement on every draft judgment underlying Sections IX/ X (Tier-1 headline verdicts are exempt – they are independent organizations’ own published verdicts).
- The new validator check’s 40% recall figure is measured on the same sample it was calibrated against – likely optimistic about generalization (Section IX).
- The vision-context fix’s effect on headline accuracy is not independently re-verified – stated as an expectation, not a measured result.
- item-0003 is permanently excluded, not scored as a failure – confirmed unfetchable across three attempts.
- Baseline comparison isolates architecture from model, but not from claim-extraction coverage – Baselines 1–3 receive TruthLens’s own already-extracted claim, so the headline comparison cannot fully separate architecture underperformance from claim-extraction giving it nothing to work with.
- The primary-source-retrieval fix is measured on tier classification, not full-text completeness, on a distinct query sample; the .gov.in fetch-reliability gap it surfaced remains open.
- RQ5 and RQ6 are deferred by design, not omission, per pre-registered fallback rules (Section XI).
- Ground truth is sourced from a small set of professional fact-checkers whose editorial judgment we do not independently audit – a systematic coverage or resolution asymmetry would propagate undetected, since no second ground-truth source exists to check against.
- TruthLens-202’s claim/label annotations are a single LLM-assisted reviewer’s work, not verified by a second human (Section VI-F). Verdicts trace to an independent professional fact-check via a recorded verbatim quote, which bounds the risk, but the claim phrasing and label mapping are unaudited – a circularity concern given that an LLM-based system is the intended evaluatee. The Section VII-D result is reported with this caveat attached and as a floor, not a validated headline; a second-annotator pass is the top open item (Section XVI).

TABLE XVII
Per-item conditions, corrected comparison ($n = 6$)

Item	Multi-claim?	Extracted?	TL vs. B2
item-0001	Yes (4)	Yes	Both correct
item-0002	Yes (2)	Yes	Both correct
item-0004	No (1)	Yes	B2 only correct
item-0005	Yes (2)	Yes	Both wrong
item-0006	No (0)	No	Both wrong; TL abstains
item-0007	No (0)	No	Both wrong; TL abstains

- TruthLens-202 is $\sim 88\%$ single-source (Alt News) and carries three domain-driven skews (83% FALSE, 80% X-sourced, 80% provenance-type). The Section VII-D evaluation therefore leads with balanced accuracy and per-class F1, not raw accuracy, and treats the source concentration as a coverage limitation, not a representative sample of the fact-checking ecosystem.

XV. Discussion

Audit your baselines as skeptically as your own system. This program’s most consequential finding came not from improving TruthLens but from a post-hoc audit that treated our own experimental design with the skepticism this paper applies to the system under test, finding the baselines quietly advantaged program-wide (Section VII-A). Without it, this paper would have published a confident, wrong-in-degree explanation for why TruthLens loses to a single-shot baseline: the diagnosed causes (claim-extraction coverage, validator false negatives) were real, but the uncorrected number overstated their effect because part of it was an artefact of unequal claim input. A baseline is part of the apparatus, not a neutral reference, and deserves the same implementation-level scrutiny as the system it is compared against.

Label accuracy vs. support validity. Two general, ground-truth-independent fixes moved validator recall $16.7\% \rightarrow 40\%$ while reel-level accuracy stayed at 2/6, before and after the baseline correction. Label accuracy and support validity are different axes that can move independently – a benchmark reporting only label accuracy would have shown no progress during a pass that materially reduced how often the system publishes a confidently wrong, unsupported verdict.

When TruthLens helps, specifically. Once claim input is held constant, full TruthLens beats two of three baselines and trails the third by a much smaller margin; the two remaining wrong items are a claim-extraction coverage failure (0006/0007) and one validator false negative (0004). Table XVII replaces a win/lose framing with where the advantage materialises.

Reading the matrix rather than the aggregate: TruthLens and Baseline 2 agree on every item where extraction succeeded and produced more than one claim (0001, 0002) – exactly where Section VIII-A’s counterfactual shows aggregation adding value. The one exclusive loss (0004) is single-claim, where decomposition contributes nothing

and the already-diagnosed validator false negative costs it. Both systems fail identically when extraction fails (0006/0007). Stated as a hypothesis for a larger dataset, not a conclusion this n supports: TruthLens’s advantage over a single-shot baseline may be concentrated in multi-claim items with successful extraction and structurally absent where extraction fails or a post has only one claim.

This tempers how DeVerna et al.’s finding [10] that curated context matters more than reasoning ability should be read against ours: the corrected comparison is broadly consistent with theirs – TruthLens’s curation-heavy architecture is competitive with, and mostly ahead of, simpler alternatives once claim input is fairly controlled – but the lesson we draw is methodological as much as architectural: curation-heavy systems should be evaluated at the component level, with baselines audited as carefully as the system, because an end-to-end label-accuracy number can misattribute a claim-input confound to an architectural one. We did not tune any prompt, threshold, or model choice on the frozen held-out set; the validator fixes and the baseline correction are all ground-truth-independent. We would rather report a real 33.3%, a real correction, and an honest account of both than a smoother number whose provenance we could not defend.

XVI. Future Work

Items already acted on in this program are marked; the rest are open.

- 1) Evaluate TruthLens-202 beyond the local-only floor (corpus built and evaluated once). Re-run the same 193-item held-out split under a larger local model, a vision-first claim-extraction path for provenance posts (where the visual is the claim and text-only extraction fails on $\sim 65\%$ of items), and a metered cloud-escalation tier, reported together as a capability-vs.-cost ablation. This is the top result-bearing open item.
- 2) A second, independent annotator for the 165 reviewed TruthLens-202 claim/label annotations (currently one LLM-assisted reviewer) and the Tier-2 draft judgments underlying Sections IX/X – the prerequisite before the local-only result is a validated headline rather than a floor.
- 3) Close claim-extraction coverage failures on degraded audio and, more broadly, the near-zero verbatim source_quote population (attacked: a strengthened prompt did not move the rate, $n = 8$; a structural change such as eliciting the quote before the claim is the next candidate). EXP-016 re-confirmed the degraded-audio gap across all 8 modality combinations.
- 4) Video-provenance / cross-post verification (stage 1 built: a perceptual-hash first-pass filter validated on real frames). Stage 2 (claim-embedding clustering) is blocked by a confirmed infrastructure gap – the local Ollama server lacks embedding support and

no embedding model is pulled; stage 3 and pipeline integration are undone.

- 5) Matched-pair sourcing for RQ5 and a large-enough set for RQ6: both remain deferred by pre-registered rule – RQ5 needs a sourcing pass matched on topic/structure/difficulty across political actors; RQ6’s binning gate (no bin under 3 items) is unmet.
- 6) Independently validate the deterministic checks against phrasings and real production disagreement cases they were not calibrated on – Checks 6/7/8 were held to a synthetic bar; the original four checks’ circularity is still open, as is the .gov.in full-page-fetch reliability gap (403s, TLS failures).
- 7) Extend the adversarial suite to the categories needing real AI-generated-image, edited-video, and audio-corruption content (prompt-injection and contradictory-sources are now covered).
- 8) Isolated test infrastructure for quota-dependent tests: the collision between real concurrent Gemini usage and mocked scenarios is quantified (a confirmed 20-requests/day cap) but not fixed, and the live publishing-account connection and everything downstream of it remain unattempted.

XVII. Conclusion

This paper reports the first evaluation of TruthLens against independent, held-out ground truth, and leads with the least flattering result it produced – one in our own experimental design, not the system. A skeptical post-hoc audit found our baselines had been given a cleaner claim input than TruthLens’s own extraction ever produced, program-wide, contradicting our stated methodology and inflating baseline performance; a research program that audits the system under test but never its own baselines is not applying its standard uniformly. Corrected, full TruthLens beats two of three baselines outright and trails the third by a substantially smaller margin on the paired six-item sample, with the residual gap tracing to claim-extraction coverage failures on two items and one validator false negative – more useful than the original partly-artifactual negative number or an uncritical positive one. A claim-decomposition counterfactual ($n = 4$) separately shows aggregation delivering a real gain (50.0% vs. 25.0%) once extraction produces more than one claim, and two ground-truth-independent validator fixes moved recall 16.7%→40% (100% precision) while leaving reel-level accuracy unchanged – a concrete demonstration that label accuracy and the formally-defined support validity construct are separable properties. RQ5 (bias) and RQ6 (calibration) are deferred, not forced, per pre-registered rules. The 24-entry taxonomy of failure modes (Appendix A) – over a third found during this evaluation, all but three paired with a deterministic check and a regression test – is as practically useful as any single number, since most, including the baseline confound, are

discoverable only by running a complete system against real content.

A subsequent foundation-phase program (Section XII) extends this without touching the frozen comparison: the set grew to 15 items; two deterministic validator checks (temporal, entity), each integrated only after clearing an evaluation bar, moved adversarial-benchmark precision/recall 81.8%/50.0%→87.5%/60.9% and closed a previously-named gap case; four independent experiments converged on the same finding – raw local-model output on real content is substantively empty or unusable a large fraction of the time, and the escalation defenses carry more of the reliability burden than a pass/fail reading suggests. A prompt-injection defense with zero prior test coverage was measured live (1/5 succeeded, 0/5 after a hardened prompt plus a deterministic delimiter fix, with the downstream validator independently neutralising the one success), and a genuinely new verdict-stage failure – label selection not weighting source reliability under contradiction (0/14 trials) – was reported unresolved after an insufficient prompt fix, then addressed with a deterministic Check 8. A benchmark-scaling session measured a structural ceiling: roughly one promotable item per 900 candidates under an Instagram-only constraint, an order of magnitude below the initial estimate, because most of this pattern’s false claims are posted elsewhere (often X/Twitter) than the fact-checked Instagram post.

Acting on that ceiling, a later pass relaxed the platform scope and added a review-gated annotation step, producing TruthLens-202 (193 validation items), and we run the full pipeline over all 193 once in a \$0 local-only configuration (Section VII-D), reported as a floor with its limitations stated rather than softened: 34.7% of posts reach a verdict, bucketed accuracy on those is 29.9% (below an always-FALSE baseline on balanced accuracy), the annotations are one LLM-assisted reviewer’s work pending a second annotator, and the corpus carries a single-source majority and three domain-driven skews. Its one firm, transferable finding is diagnostic: with a local 3B model, claim extraction – not retrieval (0% infrastructure failure) and not verdict reasoning – determines whether short-form political misinformation gets checked at all, and it fails on roughly two-thirds of real posts. Work is paused here; the immediate open items are the larger-local-model / vision-first / metered-cloud ablation on the same held-out split, a second independent annotator, and a broader ground-truth base (Section XVI), alongside the still-deferred RQ5/RQ6, the cross-post attribution problem, and the live publishing-account connection – none hidden in the sections above.

Appendix

Section IV-D2 summarizes verification-gated model cascading; full detail follows. Five stages – claim extraction, evidence analysis, vision-context, content generation, and verdict proposal – each run a deterministic substantiveness

check before accepting local-model output: claim extraction requires at least half of extracted verifiable claims to have a `source_quote` substring-grounded in the transcript/OCR, and at least one claim with non-empty text (zero claims is a legitimate “found nothing,” not a failure); evidence analysis requires a non-empty per-source explanation (firing per-source since this stage calls the LLM per source); vision-context checks `scene_description`’s word-overlap against the system prompt (the 0.20 threshold from Section IV-D2); content generation/verdict reasoning requires at least six real words after stripping citation markup. On failure, the call retries once against a paid model (Gemini) with an identical prompt, logged either way. Illustrative development-time telemetry (predating the frozen evaluation, not part of it) found this gate firing on 3/7 claim-extraction, 2/5 content-generation, and 2/8 verdict-proposal completions – small counts, large enough to show it firing routinely.

Section IV-D3 summarizes deterministic aggregation and grounded corrections; full detail follows. The reel-level rule table: all TRUE yields overall TRUE; a mix of TRUE/MOSTLY_TRUE and FALSE yields MISLEADING. Because a bare FALSE/UNVERIFIED is a weak deliverable, the verdict-proposal call also carries optional `corrected_fact` and `context_note` fields, held to the same evidence-grounding rule as `reasoning_summary`, left null rather than guessed when unestablished. We deliberately did not build the natural next step – speculating about a claim’s downstream consequences – since that is ungrounded generation of exactly the kind Section IV-D1 exists to prevent. Both fields are invalidated whenever the parent verdict fails validation, even if individually grounded.

Section X-E defines the cross-post attribution problem and states its headline number; full discussion follows. This is structural, not a sampling accident: a fact-checker traces a viral claim back to its original, differently-captioned source video, and that video’s URL is what enters the dataset (Section VI) – a majority of this dataset’s items may be structurally unwinnable for single-post claim extraction regardless of modality coverage. TruthLens has no reverse-video/cross-post matching capability and has never claimed one; we did not prototype a provenance-retrieval module against this program’s own benchmark, since evaluating it on the same 6 items already read would violate our own held-out discipline – a frozen, disjoint pilot set is the correct next step (Section XVI), not attempted here.

Section XIV lists twelve threats at a checkable density; the additional reasoning for the first ten follows in the same order (the last two, on the TruthLens-202 corpus and its local-only evaluation, are detailed in Sections VI-F and VII-D).

Small, non-independent sample. $n = 9$ (6 resolved from every configuration) is far below this program’s own 20–30 target and an ideal 50–100. Every proportion carries

an exact count and a Wilson interval; several, including the 33.3% headline, have intervals wide enough to be consistent with a different true rate.

Single domain, single country. Indian political short-form video with English/Hindi/Urdu transcripts; source-tiering domain lists and research-planning prompts are untuned and untested for other national contexts.

Single annotator, no IAA. Every draft judgment in Sections IX/X (validator TP/FP/TN/FN, evidence relevance, claim-coverage matching) is one unadjudicated reviewer’s; Tier-1 headline verdicts are exempt (they are independent organisations’ published verdicts).

The new validator check’s 40% recall is measured on its own calibration sample. Likely optimistic for differently-phrased “no evidence found” reasoning, since the phrase list was written after reading the two cases it now catches; the principle is ground-truth-independent, the implementation in this narrow sense is not.

The vision-context fix’s headline effect is not independently re-verified – stated as an expectation (root causes of items 0006/0007 diagnosed as unrelated to vision quality), not a measured result; the paid fallback’s quota was exhausted before an end-to-end test.

item-0003 is permanently excluded, not scored as a failure – confirmed unfetchable across three attempts, a real content-acquisition limitation against Instagram, unaddressed by any fix here.

The baseline comparison isolates architecture from model but not from claim-extraction coverage. Baselines 1–3 receive TruthLens’s own extracted claim, so the headline comparison cannot separate “downstream architecture underperforms” from “extraction gave it nothing for 2 of 6 items”; Section VII argues for the latter.

The primary-source-retrieval fix is measured on tier classification, not full-text completeness, on a query sample distinct from the main evidence-quality run; the .gov.in fetch gap it surfaced (403s, TLS) is open.

RQ5 and RQ6 are deferred by design, per pre-registered fallback rules (Section XI), not reported with an underpowered number.

Ground truth is a small set of professional fact-checkers whose editorial judgment we do not independently audit. Using their published verdicts as Tier-1 truth is standard, but a systematic asymmetry in which claims they cover or how they resolve ambiguity would propagate into every number here undetected, with no second ground-truth source to check against.

Section XI states the headline RQ5/RQ6/efficiency facts; full discussion follows.

RQ5 qualitative observation, in full. Of the 9 real verdicts audited in Section IX, a verdict targeting a government body’s own claim (item-0004’s Delhi Police baton case) was treated no differently in kind by the pipeline than verdicts about party or activist claims – the system was willing to reach FALSE against a claim traceable to a government actor’s own public position (a

verdict itself judged wrong for reasons unrelated to which actor was involved). One data point showing the system does not categorically refuse to find against a government actor.

RQ6 caveated observation, in full. The two highest-confidence verdicts (0.7, 0.8) are both cases Section IX’s draft human review judged unreliable or factually wrong, while several of the lowest-confidence verdicts (0.0–0.1) were appropriately cautious UNVERIFIED outputs. At $n = 9$ this is not evidence of an inverse relationship between confidence and correctness – it is exactly the kind of pattern a real calibration analysis, at real scale, would need to check for rather than assume away.

Efficiency gaps, in full. Baseline 4 (TruthLens minus validation) has an identical call-count/token/latency profile to full TruthLens by construction, since SKIP_VALIDATION changes only which content gets persisted, not which LLM calls get made; no separate timed pass was run for it. Baseline 1 (LLM-only) reuses timing from an earlier script that also did not capture latency, a gap inherited rather than newly introduced here. Exact dollar cost per configuration (Gemini pricing was not itemized per call during this run) is a concrete, scoped gap for a future pass.

This program’s earlier “each configuration’s own full $n = 9$ run” table (Section VII), retained only for the historical record and not cited as evidence for any claim in this paper: 3 of its 9 rows (items 0003/0008/0009) still use the uncorrected claim-input method, since none has both a complete real extraction and a complete real full-TruthLens verdict to correct against (Section VII-A).

TABLE XVIII

Superseded: each configuration’s own full run, $n = 9$ (not used for any claim in this paper)

Config	n	Correct	Acc.	95% CI
B1: LLM-only	9	2	22.2%	[6.3,54.7]
B2: Search+LLM	9	7	77.8%	[45.3,93.7]
B3: Search+RAG+LLM	9	5	55.6%	[26.7,81.1]
Full TruthLens	6	2	33.3%	[9.7,70.0]

Section IX summarizes the four-way primary-source metric and three further findings from the evidence-quality audit; full detail follows.

The four novel paraphrase test cases (Check 4 generalization test). “the data does not corroborate this,” “insufficient corroboration exists,” “the available sources do not establish,” and “cannot verify this claim” – none in the original phrase list, each paired with a confident label in a synthetic but realistic evidence matrix; all four were missed.

The four-way primary-source metric, per-metric detail. (1) Source-tier classification rate: $16/68 = 23.5\%$. (2) Relevant primary source rate (draft human judgment): $11/16$ (68.75%) of primary-tier sources are topically relevant; the 5 not relevant include a wrong-domain case (engineering-admissions authority for a medical/NEET claim) and a

wrong-country case (US National Archives for an Indian claim). (3) Primary source fetch-success rate: $16/16$ by construction of an existing guarantee that the pipeline never stores an unfetched source; full-vs-snippet text was not separately checked, a gap shared with the search-side fix below. (4) Usable evidence extraction rate: $3/16 = 18.75\%$ (stance $\neq$ irrelevant).

A second major bug: empty explanations in 47.1% of evidence rows. A real government source was classified contradicts with zero explanation text – no way to audit why. Checking systematically: $32/68$ (47.1%) real evidence rows had an empty explanation field, the same schema-valid-but-empty class found elsewhere, never guarded in this stage. Fixed with a per-source substantiveness check (Section IV-D2); live re-verified against the exact case that surfaced it, producing the same stance with a real, specific explanation. The deeper issue this reveals is less “the stance was wrong” and more that claim extraction had produced a claim general enough that unrelated-event sources become plausible-looking evidence for it – a claim-specificity problem as much as an evidence-relevance one.

Closing the primary-source gap: a deterministic search-side fix. A research-planning query targeted at primary sources is restricted at the search-call level to a fixed government/legal domain allow-list, rather than trusting a free-text site:gov.in the model writes into its own query. A related source-classifier gap (missing NIC-/RBI-hosted domain patterns) was fixed alongside it. Re-running seven real, unscripted domain-restricted queries: four returned results, pooling to 20 sources, 19 of which (95%) landed in a primary tier, against an $\sim 11\%$ (8/72) baseline before the fix. Two findings temper this: domain restriction guarantees tier correctness, not topical relevance (one query returned five correctly-tiered but topically irrelevant results, the same pattern as the engineering/NEET and US/Indian mismatches in Section IX); and for one query’s five URLs, every full-page fetch failed (403s, TLS errors), falling back to snippet text – manually confirmed substantive, but this .gov.in fetch-reliability gap remains open, not folded into the 95% headline figure.

A prototype entity-consistency check, evaluated honestly and not integrated. Motivated by Section XIII’s entity-confusion failure mode, we prototyped a deterministic check: does at least one of a claim’s extracted entities appear in each non-irrelevant cited evidence row’s text? Evaluated against the same real 9-claim sample (ENTITY_CONSISTENCY_EVALUATION.md), 7/9 evaluable rows were flagged; manual audit found one genuine true positive (a Delhi -Police claim’s “contradicting” evidence was actually about an unrelated Burdwan, West Bengal incident, refining item-0004’s error attribution), four false positives from a fixable gap (the claim’s only entity was the abstract concept “Democracy,” not a person/organization/location), and one likely false positive from transliteration variance (“Dipke”/“Deepke”). The historical case that most

motivated this check – a claim naming no one, whose evidence smuggled in an unestablished identity – has no entities to check and this design cannot catch it. Not integrated into production: one true positive on ~5 evaluable cases is too small to justify a fifth check without more data (Rule 4). Reported as an honest, mixed result, not suppressed or overclaimed.

Section XIII summarizes this taxonomy. All 24 entries are listed below, one line each, so none is dropped for space; several (entries 23–24) are the load-bearing measured results discussed in the body.

- 1) Schema-valid, substantively empty output. A local model emits a schema-conforming but information-free string; found independently in four stages; each needs its own content-level substantiveness check (Section IV-D2).
- 2) Verbatim-but-misattributed quotation. A source quote is a real on-screen substring but is the publisher’s caption about a person, not their words; fixed by rejecting a quote that contains the candidate speaker’s own name.
- 3) Silent verdict/presentation inconsistency. A dashboard showed the primary claim’s verdict while the published output showed the reel-level verdict; the two could disagree with no error; found only by diffing rendered output against the database.
- 4) Bracket-matching truncation bug. A non-greedy sanitization regex stopped at a bracket inside a list-repr in the text it was stripping, leaving an internal note on a slide; fixed by truncating at a literal marker.
- 5) Unrepresentative thumbnail selection. One fixed-offset frame is unreliable; now several frames are sampled and picked by grayscale-variance complexity to skip blank/transition frames.
- 6) Entity confusion between similarly-named organizations. Retrieval returned an article about a different same-named org; evidence analysis labelled it contradicts rather than irrelevant; not currently checked.
- 7) Downgraded reasoning reused downstream. A validator forces a verdict’s label but not its free text; that text (carrying an entity-confusion fabrication) was reused verbatim; fixed by disqualifying any non-passed verdict’s text from reuse.
- 8) Single-digit fabrication in generated headlines. A headline prompt forbidding new numbers turned “4,000 crore” into “\$1 Billion”; the number-grounding check skips single digits; fixed with a headline-specific digit-grounding check that falls back to the claim’s own text.
- 9) Vision-read text silently discarded. Vision read a figure/timeframe into a dedicated “visible text” field that claim extraction’s prompt never included; fixed by weighting it like OCR.
- 10) Missing glyph rendering as a blank box. Bundled fonts lack the Rupee sign, routine in this domain; fixed with a render-time substitution.
- 11) Vision-transcription accuracy, unresolved. Vision occasionally misreads text (truncated words, stray characters) without discarding it – an underlying-model limitation left unpatched rather than risk an unverified heuristic.
- 12) Empty-string claims passing schema validation. A non-empty-string type does not require a non-empty string; fixed with a substantiveness check plus an unconditional blank-text filter.
- 13) except against the wrong SDK exception hierarchy. A routine quota 429 raised an unrecognised type and propagated unhandled; a pre-existing test mocked the same wrong hierarchy so it never caught it; fixed against the real hierarchy, test rewritten against real instances.
- 14) Baseline silently inheriting a rescue mechanism under evaluation. The first baseline draft used the provider factory, which returns a Gemini-fallback-wrapped provider when a key is set; caught in smoke testing before any number was reported; fixed by importing the local-only provider directly.
- 15) A second “no video” phrasing masking a fetchable photo post. item-0005’s No video formats found! did not match the known photo-fallback phrase, so a fetchable post failed hard; fixed by matching a tuple of phrasings.
- 16) UUID hex fragments leaking as fake unsupported numbers. Digits inside a cited evidence ID’s UUID were read as fabricated statistics, in two citation-markup forms; the fix also removed an accidental true positive from an earlier audit, itself reported.
- 17) 47.1% empty explanations in per-source evidence analysis. The one LLM stage with no substantiveness check of its own; confirmed at scale once measured; fixed with a per-source check.
- 18) Recurring vision prompt echo, previously undercounted. The local vision model repeatedly returns a schema-valid echo of its own system prompt across ≥ 4 real reels; fixed with a word-overlap-with-prompt check.
- 19) A label contradicting the model’s own reasoning. Reasoning stating “no reliable evidence” paired with a confident non-UNVERIFIED label – a distinct category from a fabricated citation or number; fixed with a new deterministic check whose calibration circularity is disclosed as a threat.
- 20) MissingGreenlet crash from a stale ORM read after rollback. A per-claim handler’s db.rollback() expired a loaded object; the next attribute access crashed the whole script; fixed by capturing values before the try block.
- 21) A research script’s stdout and its JSON summary silently disagreed. day8_final_tables.py reused k/n across three computations; the end-of-function JSON combined two tables’ counts into a fabricated-

looking $n=4$, correct=3 (75%) matching neither printed table nor the published 33.3%; caught only by visually checking the figure against its table; fixed with distinct variable names and explicit table persistence.

- 22) A daily API quota colliding with mocked tests. A quota-aware cooldown check reads the same shared table real usage writes; concurrent real exhaustion short-circuits mocked scenarios into the cooldown path; diagnosed to a confirmed 20-requests/day free-tier cap; not fixed (needs test isolation), worked around by re-running after real usage finished.
- 23) A structural gap in the delimiter prompt-injection defence. The data-wrapping convention never neutralised literal delimiter tokens already in the untrusted text, allowing a forged boundary; fixed deterministically and regression-tested both directions. Separately, 1/5 live injection attempts (a blunt override) succeeded pre-fix, 0/5 after a hardened prompt; carried through the pipeline, the downstream validator independently resolved the one success to a correct FALSE.
- 24) Verdict-label selection does not weight source reliability under contradiction. A 0.95 primary source supporting vs. a 0.20 uncited source contradicting produced no TRUE/MOSTLY_TRUE in 0/14 trials, some with reasoning that named the weak source as unreliable yet did not act on it; a hardened prompt did not fix it (0/7); a deterministic Check 8 (cited-reliability vs. verdict direction) was built, cleared the same bar as Checks 6/7, and integrated with zero measured regression.

Full numeric detail for Section XII; every number traces to experiments/registry.jsonl and a dated artifact under research/results/.

- EXP-013/019 (validator Checks 6–7). Temporal and entity checks, 7+7 regression tests, then the 34-case adversarial benchmark jointly: TP 14, FP 2, FN 9, TN 9 (precision 87.5%, recall 60.9%). Both FPs are a pre-existing Check-3 contrastive-reasoning limitation. A live end-to-end test was inconclusive (an earlier check fired first); Check 6’s logic is covered by unit tests.
- EXP-012/016 (source-quote groundedness). Across 8 DEV items (12 claims) and again at $n=90$ (8 items $\times$ 8 modality combinations), 0 claims carried a verbatim-grounded source_quote; spot-checked as real behaviour, not a measurement artefact.
- EXP-016 (8-combination coverage). Ground-truth claim coverage by modality: caption/audio/OCR 25.0%, vision 37.5%, OCR+vision 12.5%, audio+OCR+vision 25.0%. Two items produced 0 claims in all 8 conditions (including the one TRUE item); a third produced 5 claims, none covering the GT assertion (a wrong-claim failure).
- EXP-015 (5-query retrieval). 4 DEV claims, 20

queries, 42 sources, 16 primary-tier. Relevant-primary rate by query type: primary_source 3/4, context_history 2/2, contradiction 1/3, entity_focused 1/4, exact_claim 0/3 – the last driven by one claim whose 7 primary-tier hits were all keyword-collision matches on unrelated .gov.in domains (domain-pattern tiering has no content-relevance check).

- EXP-017 (evidence-stance taxonomy). An 8-category candidate taxonomy vs. 20 real rows: raw change 55%, but 80% had an empty explanation on one side (no substantiveness-retry safeguard); filtered to 4 genuinely substantive rows, 1 changed – n too small. The comparison’s own confound is the more general finding.
- EXP-018 (cross-post detection). Same-video control: minimum Hamming distance 0; different-video control: 20 (threshold 10). A genuine independently-posted-same-footage pair (item-0002) could not have its post URL extracted from the citing article – not achieved this pass.
- EXP-020 (adversarial stress test). Six synthetic cases through the real claim extractor: garbled_ocr 0, caption/transcript contradiction 1 (caption only), extremely_short 0, mixed_language 2 (English portion only), repetitive_spam 1 (40 reps collapsed), near_max_tokens 0 (confounded by a Gemini quota event). Zero crashes.
- EXP-021 (source_quote prompt strengthening). A v4 candidate with a 4-step self-check + worked example vs. v3 on the same 8 items: v3 9 claims / 0 quoted, v4 6 claims / 0 quoted; on the item the example was built from, v4 produced 0 claims – the stronger instruction did not recover the quote and may have suppressed extraction.
- EXP-022 (input-signal consistency feasibility). Jacard of caption vs. transcript keywords for the 4 reels with both fields: 0.000 in all 4 – the normal case for legitimate multilingual content, not inconsistency.
- EXP-023 (multilingual bias re-examination). Script classification of all 90 EXP-016 claims: 42 Latin, 48 non-Latin; a Marathi item 37 non-Latin / 8 Latin (the Latin ones caption metadata), a Hindi item 5/5 non-Latin, English items Latin-only, the one code-switched item 7 Latin / 6 non-Latin – direct evidence against a general Latin-script bias.
- EXP-024 (VALIDATION-split real population). The real second half of analyze_reel() run and committed for all 6 VALIDATION items: 4 produced 0 claims; item-0010 2 claims (one entity_mismatch downgrade, one FALSE/passed); item-0014 1 claim (FALSE/passed, matching ground truth). The entity-mismatch case audited as a genuine correct catch – Check 7 generalising to real data it was never calibrated on.
- EXP-025 (aggregation counterfactual on real VALIDATION data). Only $n=1$ VALIDATION item has

- ≥ 2 verdicted verifiable claims (item-0010); single-claim-only misses, full `derive_overall_verdict()` decomposition resolves to FALSE – directionally consistent with, not confirmation of, the $n=4$ DEV result.
- EXP-026/027 (prompt-injection stress test). 5 synthetic reels, one injection style per field: pre-fix (v3) 1/5 succeeded (direct override), post-fix (delimiter sanitisation + hardened v4) 0/5. The one pre-fix success, carried through the real pipeline, reached FALSE/passed with evidence-grounded reasoning.
 - EXP-028 (evidence-stance re-run with the production safeguard). As EXP-017 but with the substantiveness-retry active: both attempts hit Gemini cooldown/429 (0/20, then blocked), clean rows 6/20 then 5/20; 1 reclassification, assessed as a likely taxonomy-misapplication, not counted as a positive.
 - EXP-029 (contradictory-sources verdict stress test). The `reliability_weighted_conflict` case (0.95 government support vs. 0.20 uncited contradiction), re-run for stability: pre-fix (v2, $n=7$) MOSTLY_FALSE $\times 3$, UNVERIFIED $\times 3$, OUTDATED $\times 1$; post-fix (v3, $n=7$, explicit reliability rule) UNVERIFIED $\times 5$, MOSTLY_FALSE $\times 2$. 0/14 produced TRUE/MOSTLY_TRUE; a `majority_with_credible_outlier` sanity check showed similar post-fix instability.
 - EXP-030 (Check 8 evaluation). A deterministic reliability-direction check (flags when the max-cited-reliability gap > 0.4 and the verdict sides with the weaker stance): 10/10 synthetic cases; replaying EXP-029’s 19 real labels, 14/14 `reliability_weighted_conflict` flagged, 0/5 `majority_with_credible_outlier` false-flagged; 0 interactions with the 34-case benchmark. Integrated as Check 8 with identical pre/post precision/recall (87.5%/60.9%).

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